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Competition

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Trading Fees and Intermarket Competition

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Abstract

We study the 2013 changes in maker-taker pricing fees implemented by BATS on its two European venues, CXE and BXE. The CXE rebate reduction deteriorates market quality and market share, whereas the BXE rebate removal and take-fee reduction improve them. We derive a model of two competing limit order books, in which large (small) stocks are characterized by investors with higher (lower) propensity to supply liquidity and by greater (lower) trading activity. Consistent with our model, we show that traders in large stocks are more reactive to rebate reductions while traders in small stocks are more reactive to take-fee reductions.

JEL Classifications: G10, G12, G14, G18, G20, D40, D47

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1. Introduction

In today’s fragmented equity trading environment, trading venues use trading fees to compete for order flow. Most venues operate limit order books, and rely on endogenous provision of liquidity, therefore many of them subsidize liquidity supply by offering a rebate (a negative make fee) to traders submitting limit orders. However, trading venues have to generate revenues to cover their costs and therefore impose a higher positive fee (take fee) on market orders.¹ This type of pricing, called maker-taker pricing, is actively debated among academics, practitioners, market operators, and regulators. Maker-taker pricing is an important competitive tool for trading venues in today’s fragmented markets, and may benefit investors to the extent that it allows intra-tick trading, thus reducing the trading frictions caused by the fact that prices are discrete. However, maker-taker pricing has recently been criticized for potentially exacerbating conflicts of interest between brokers and their customers, for contributing to market fragmentation and market complexity, and for undermining price transparency.²

Resolving this debate requires a fundamental understanding of the effects that maker-taker pricing has on market quality and on the distribution of order flow across venues. To identify these effects absent a randomized control trial or natural experiment, we rely on venue-specific changes in maker-taker pricing as the treatment, and compare changes in market quality on, and market share of, treated and competing venues in the same market, to changes in the same measures for a control sample from a different market. Specifically, we investigate the impact of fee changes on the trading behavior of investors in a multi-venue framework. We study the January 2013 change to maker-taker pricing implemented by BATS Europe (BATS) in its European venues, BXE and CXE.³ BATS eliminated the rebate on liquidity supply and reduced the positive charge on liquidity demand on BXE, and reduced the rebate on liquidity supply on CXE.⁴ Meanwhile, the maker-taker pricing fees charged by its main competitor, Turquoise (TQ), remained unchanged. We examine the effects of these fee changes on displayed liquidity (market quality) and market share using a difference-in-difference approach, drawing a sample from the Australian stock exchange to control for global changes in liquidity.

To derive empirical predictions, we develop a model of two identical limit order books with

¹According to the [OICV-IOSCO \(2013\)](#) report, there exists at least four types of fee structures: the symmetrical pricing model, with both the active and passive side of a trade paying the same fee; the asymmetrical pricing model, with both the active and the passive side of a trade paying a fee, but the fee paid is not necessarily the same; the maker-taker pricing model, with the provider of liquidity (maker) receiving a rebate and the taker of liquidity (taker) paying a fee; and the inverted maker-taker pricing model, with the provider of liquidity paying a fee and the taker of liquidity receiving the rebate.

²For extensive background and critical review on access fees, see the SEC Market Structure Advisory Committee’s October 20, 2015, Memorandum “Maker-Taker Fees on Equities Exchanges.”

³BATS Europe is a subsidiary of BATS Global Markets, which in turn was acquired by CBOE Holdings in March, 2017.

⁴Note that consistent with [Chao, Yao, and Ye \(2018\)](#), traders’ heterogeneity leads to multiple fee structures set by BATS for its trading platforms.

discrete prices - i.e., governed by a tick size - that compete for the provision of liquidity. Our model extends the existing literature by providing analytical solution to a model with two competing standard limit order books, discrete prices and trading fees. As predicted by the model, we document significant changes in market shares and displayed liquidity following the fee changes, both for BATS' venues and for Turquoise. Moreover, we discover significant cross-sectional differences in the response to fee changes.

Maker-taker pricing has enabled new entrants to compete effectively with incumbent exchanges, potentially leading to narrower quoted spreads, but the practice has been also criticized. [Angel, Harris, and Spatt \(2015\)](#) argue that maker-taker pricing obfuscates true spreads, that it distorts order routing decisions, and that it hurts both internalizing dealers and venues that do not use maker-taker pricing.⁵ [Harris \(2013\)](#) further argues that rebates allow traders to circumvent the minimum price variation (tick size), thus by-passing Regulation NMS order protection rules. [Angel et al. \(2015\)](#) recommend that the SEC either requires that all brokers pass through access fees and liquidity rebates to their clients and clarify that best execution obligations apply to net prices instead of quoted prices, or prohibit maker-taker pricing altogether.

On the other hand, [Malinova, Park, and Riordan \(2018\)](#) see no reason to abolish maker-taker pricing as academic evidence suggest that HFTs and other traders pass through a significant fraction of the rebates to active traders.⁶ Instead, they support initiatives to provide investors with better information about execution quality that includes maker-taker fees. [Foucault \(2012\)](#) shows that the make-take fee breakdown can affect the mix of market and limit orders and may even increase market participants' welfare. Consequently, he advocates that exchanges and regulators conduct pilot experiments to assess the effect of maker-taker fees on the composition of order flow (market vs. limit orders) before contemplating any changes to the current rules.

Not surprisingly, industry participants, exchanges and the SEC have also weighed in on the maker-taker pricing debate. Some argue that the maker-taker pricing contributes to market complexity, and that the SEC should either eliminate maker-taker pricing or at least lower the cap on access fees. Others claim markets are functioning well and that no changes to fee caps are needed. In March 2018, the SEC proposed a Transaction Fee Pilot for NMS stocks (Rule 610T) with the goal to "facilitate an informed, data-driven discussion about transaction fees and rebates and their impact on order routing behavior, execution quality and market quality in general" according to SEC Chairman Jay Clayton.⁷ Rule 610T would create two

⁵This concern has been validated using options market data, [Battalio, Corwin, and Jennings \(2016\)](#) who show that retail brokers appear to route orders to maximize order flow payments: selling market orders and sending limit orders to the venues paying large liquidity rebates, and that retail traders limit order execution quality is negatively related to the level of the liquidity rebates.

⁶[Hendershott and Riordan \(2013\)](#) also show that high frequency traders (HFT) market makers pass through some of the rebates to active traders.

⁷<https://www.sec.gov/news/press-release/2018-43>.

test groups of stocks: one with a new lower cap of \$0.0010 on positive make and take fees (but no cap on rebates); and another that prohibits rebates but keeps the current fee cap of \$0.0030. The remaining stocks would be control stocks and the current fee cap of \$0.0030 would apply and there would be no restrictions on rebates. Importantly, only exchanges would be subject to the rule, but not Alternative Trading Systems (ATS). Assuming no cross-subsidization across stocks, a maker-taker venue would need to lower both the rebate (so that it is lower than the take fee) and the take fee to meet the lower cap for stocks in the first test group under the Rule. For stocks in the second test group, rebates would have to be lowered to zero, while the take fee in principle could remain unchanged for a maker-taker venue under the Rule. However, competitive pressures would in the latter case likely result in a lowering also of the positive take fee. The proposed Transaction Fee Pilot was adopted by the SEC in December 2018. However, ISE, Nasdaq, and Cboe together filed a petition challenging the Pilot, arguing it was arbitrary and capricious, would give other market players not subject to the Rule an advantage, and that the Pilot was beyond the SEC’s authority.⁸ A federal appeals court in June 2020 sided with the petitioners, concluded that the regulator exceeded its authority, vacating the Rule and remanding the case.⁹

Therefore, we need to look to markets outside of the U.S. to gather evidence on how maker-taker pricing affects displayed liquidity and the distribution of order flow across venues. That is exactly what we do in this paper. Like the envisioned Transactions Fee Pilot, we study a fragmented market where different fee structures apply in different venues for the same stock. However, unlike the Pilot, the fee reductions we study are voluntary and are undertaken by Multilateral Trading Facilities (MTFs) - the European equivalent of ATSs - instead of by an exchange. The CXE rebate reduction maps into Rule 610T test group one, whereas the BXE elimination of the rebate and reduction of the take fee maps into what we would expect to happen under Rule 610T for test group two.

To help us frame the empirical analysis, we develop a model of a dynamic limit order book with a discrete pricing grid that faces competition from another identical limit order book. Our model departs from Buti, Rindi, and Werner (2017) in that it has endogenous liquidity supply, trading fees and a competing limit order book.¹⁰ We use the model to derive predictions on the effects of a change in fees on market quality and market share in a fragmented market. The new feature of our model is that it includes both the tick size and a competing venue. Our model complements both the Colliard and Foucault (2012) model in that it has a tick size, and the Foucault, Kadan, and Kandel (2013) model in that it includes a competing market.¹¹

⁸Only one national exchange, IEX, filed an amicus brief siding with the SEC.

⁹<https://www.sec.gov/news/press-release/2018-298>, <http://business.cch.com/srd/19-1042-1847356.pdf>

¹⁰The first version of this paper included a model of a limit order book competing with a crossing network. We thank Charles Jones, Björn Hagströmer, and Satchit Sagade for suggesting to investigate the model with two competing limit order books.

¹¹Foucault et al. (2013) focus on monitoring costs in a steady state equilibrium, while we study the effects

This is relevant as [Colliard and Foucault \(2012\)](#) show that in a market with competing venues the absence of a tick size allows investors to react to a change in trading fees by changing their order submission strategies in such a way that they neutralize the change in the total fees breakdown. [Foucault et al. \(2013\)](#) subsequently show that when the limit order book is governed by a tick size, investors may no longer neutralize the change in fee breakdown. However, [Foucault et al. \(2013\)](#) obtain this result within a single market framework. We instead show how a change in both a symmetric and an asymmetric fee breakdown affects investors' order submission strategies when a limit order book with discrete prices faces competition from another identical limit order book. Finally, the focus of our analysis is on the effects of a change in fees on the quality of the limit order book instead of the optimal fee structure as in [Chao et al. \(2018\)](#) and [Riccó, Rindi, and Seppi \(2023\)](#).

Our model is solved by backward induction to explore all possible states of the limit order book over the course of the dynamic trading game. This comprehensive approach, despite being limited to a finite number of trading periods, allows us to capture every potential transmission channel that could be activated by the investors' order submission strategies. Consequently, our model accounts for the effects on the spread not only at the Best Bid Offer (BBO) but also across different price levels. By considering every possible state of the book, characterized by varying price levels, our model can thoroughly examine how investors' strategic interactions with the limit order book affect their order submission decisions. This analysis extends beyond just the spread to include the depth (i.e., the number of shares available) at different price levels. This feature is novel in the literature and is crucial for understanding the transmission mechanisms at play, especially when assessing the impact of changes in trading fees on trading instruments with varying degrees of liquidity.¹²

With the support of our model we show how fee changes affect different metrics of market quality in a market that has a tick size and at the same time faces competition from another trading venue. We also show how fee changes on one venue in a fragmented market are likely to influence traders' decisions on where to route their orders, leading to a migration of orders between venues.

of different trading fees on the investors' dynamic order submission strategies and thereby single out the transmission channels at work.

¹²Our approach differs from that of [Goettler, Parlour, and Rajan \(2005\)](#). They numerically solve the model to achieve a Markov-perfect equilibrium. This approach requires agents to condition their strategies solely on the current order book, disregarding any previous books. Additionally, to maintain stationarity, the model assumes that the execution probability of a limit order depends on an exogenous cancellation rate. This assumption can influence investors' strategic choices between limit and market orders. In their model, if the cancellation rate is set to zero, the execution probability of a limit order becomes one, making it impossible to distinguish between market and limit orders. In contrast, our model does not aim to establish a stationary equilibrium and does not allow investors to cancel their orders. While a model with endogenous cancellation is absent from the literature and falls beyond the scope of this paper, adopting an exogenous cancellation rate similar to [Goettler et al. \(2005\)](#) would likely encourage more aggressive investor behavior. However, we speculate that this assumption would not qualitatively change our results.

More specifically, we model two identical competing limit order books, Market I and Market II, to study the effects of fee changes in Market I. We analyze how these changes impact Market I's displayed liquidity and market share, and also investigate any effects on Market II, where fees remain unchanged at zero. Our model predicts that reducing the rebate on liquidity provision (negative make fee) in Market I, without changing the take fee (similar to the CXE fee change), will lead to a migration of orders to Market II. Consequently, the model predicts a deterioration in market quality and market share in Market I, while expecting improvements in market quality and market share in Market II. Our model shows that these effects are stronger for stocks where competition for the provision of liquidity and trading activity are more intense. This strengthening amplifies the negative effect on liquidity supply (limit orders) and induces stronger negative effects on liquidity demand (market orders). Therefore, we expect that the reduction of the rebate on liquidity provision will have a stronger negative effect on large stocks, characterized by higher levels of trading activity and, consequently, greater competition for liquidity provision, compared to small stocks.

Our model also predicts the effects of a simultaneous reduction in the rebate on liquidity provision and the take fee in Market I, while trading fees remain unchanged at zero in Market II - essentially the BXE fee change. Our results show that the reduction in the take fee makes Market I cheaper, attracting orders at the inside quotes and resulting in an improvement in market quality and market share in Market I. In contrast, the reduction in rebate on liquidity provision has a counterbalancing negative impact on Market I compared to Market II. Interestingly, our findings show that for small stocks - those with lower competition in liquidity provision - the negative effect of reducing the rebate on liquidity provision is weaker, while the positive effect of reducing take fees on market quality is stronger. Conversely, for large stocks, the opposite holds true. Therefore, we can conclude that the overall effects of a simultaneous reduction in the take fee and rebate on liquidity provision are more favorable for small stocks compared to large stocks.

To provide intuition for this result, consider the dynamics between Market I and Market II: Market I operates as a maker-taker venue, offering a rebate for liquidity suppliers and charging a fee to liquidity demanders. On the other hand, Market II employs symmetric pricing, where there are no fees for either liquidity suppliers or demanders, making it a cheaper alternative. Due to the fee incentives in Market I, liquidity suppliers are motivated to participate, hoping to earn rebates. However, because Market II is fee-free and thus cheaper overall, most trading activity concentrates on the inside quotes of Market II where execution costs are lower. Nevertheless, some trading persists in Market I because patient liquidity suppliers seek rebates and place their orders at wider quotes. They hope that their limit orders will execute at better prices once the liquidity available in Market II is exhausted. In maker-taker markets like Market I, especially for large stocks with ample trading opportunities

and heightened competition for liquidity provision due to attractive rebates, more liquidity tends to accumulate beyond the inside quotes.

When both the take fee and the rebate are reduced on Market I, we observe two main channels of transmission with opposite effects: First, the reduction in the take fee induces investors to move from the inside quotes of Market II to the now cheaper inside quotes of Market I, positively impacting Market I. This effect is relatively stronger for small stocks, where there is less competition for the provision of liquidity, making the propensity of investors to demand liquidity relatively larger than the propensity to supply liquidity. Second, the reduction in the rebate reduces the incentive for liquidity suppliers to build liquidity at prices beyond the inside quotes in Market I, negatively impacting Market I. This negative effect is weaker for small stocks, where the liquidity posted beyond the first level of the book by traders seeking rebates is presumably smaller due to the lower trading activity and reduced competition for liquidity provision. We therefore obtain that overall the positive effect of a simultaneous reduction in both the take fee and the rebate is stronger for small than for large stocks.

In real markets, it is the relative fees that matter for traders' order selection and order routing decisions. Hence, when testing our model predictions, we consider the net change in trading fees. Therefore, we not only consider the direct reduction in the CXE rebate with respect to TQ, which did not change its pricing, and the direct reduction in the BXE rebate and take fee with respect to TQ, but we also consider the net reduction in BXE fees with respect to CXE, which reduced its rebate.

For our empirical analysis, we use a stratified sample of 120 LSE-listed firms and run a panel difference-in-difference regression specification, having 120 matched-firms from the Australian stock exchange as control, to uncover the effects of BATS fee changes on market quality and market share. Our results are consistent with the empirical predictions of our model. In particular, we find that the CXE's rebate reduction results in a deterioration of market quality and market share for CXE, and an improvement of market quality and market share - for large stocks - for the competing venue TQ. Recall that this is what we would expect to happen for stocks in test group one of the SEC Transactions Fee Pilot. We also find that the change in the BXE's maker-taker pricing results in improved BXE market quality and market share - stronger for small stocks - and in a deterioration of market share for the competing venue TQ in small stocks. This is what we would expect to happen for stocks in test group two of the Pilot. Hence, consistent with our model's predictions, we find that while the competing venue TQ did not change its maker-taker pricing, it was affected by both of the BATS fee changes. In particular, the overall effect on TQ market quality and market share is consistent with both order flow migration to TQ for large firms - due to CXE's fee change- and with order flow migration away from TQ for small firms - due to BXE's fee changes. Concerns

about these types of spillovers from exchanges subject to the fee caps to ATSS that would not be subject to the caps, were a major reason for the petitioners’ successful challenge to the SEC Transactions Fee Pilot.

To evaluate the robustness of our findings, we proceed as follows. First, we re-estimate our difference-in-difference specification using LSE as a control. Though we do not believe LSE is a direct competitor to BATS, we follow the methodology of [Boehmer, Jones, and Zhang \(2020\)](#) and identify the overall treatment effect (calculating both the direct effect and indirect effect via LSE) on BXE, CXE and the competing TQ market. The new results support our previous findings. Second, we look at the effect of the fee change on cum-fee spreads (quoted spread plus twice the take fee) to test whether our results on market quality are driven by traders’ neutralizing the change in fees through quote adjustments of the bid-ask spreads ([Colliard and Foucault \(2012\)](#)). We find strong effects not only on quoted spreads but also on the cum-fee spreads. Lastly, we investigate but do not find support that our cross-sectional findings - the different effects of fee changes for large and for small firms - are driven by relative tick-size. Focusing on firms that have similar levels of relative tick-size, we still find results that are qualitatively and quantitatively different for large and small firms.

Our paper contributes to the literature by taking intermarket competition into account when studying the effects of maker-taker pricing fee changes empirically. We show that both the reduction in the rebate, and the simultaneous reduction in the rebate and in the take fee have a different effect for large capitalization stocks compared to small capitalization stocks. In particular, the reduction in fees leads to worse market quality for large capitalization stocks. This finding has policy implications since - in contrast with critics of maker-taker pricing (including market operators such as BATS) who have asked for reductions or even bans of maker-taker pricing - it suggests that the elimination of rebates is going to be particularly detrimental for liquid stocks.

The paper is organized as follows. In Section 2 we briefly review the existing literature and in Section 3 we present the theoretical model and discussion of our empirical predictions. We present our data sets and the methodology in Section 4. In Section 5 we discuss our empirical results. Section 6 consists of conclusions and the policy implications of our findings.

2. Literature review

Theoretical models of maker-taker fees have initially focused on whether the breakdown of the total fee charged by a venue into rebate and take fee matters for order flow composition, market quality, and welfare. [Colliard and Foucault \(2012\)](#) develop a model of limit order trading with no tick size and show that the breakdown does not affect the order flow composition, the trading rate, or welfare. [Foucault et al. \(2013\)](#) model of a limit order book with a positive

tick size, populated by two distinct groups of algorithmic traders with monitoring costs – market makers and market takers, – and show that the total fee breakdown matters. [Brolley and Malinova \(2013\)](#) model a dealer market with informed limit order traders to show that the breakdown of the total fee matters when investors pay a flat fee while liquidity providers incur take fees and receive rebates. More recently, two papers study the optimal market access pricing. [Chao et al. \(2018\)](#) model a 2-period limit order book with a tick size to show that in equilibrium the optimal fee structure is either the maker-taker or -symmetrically- the taker-maker. They show that an exchange setting make and take fees simultaneously chooses the price of the execution service (make fee) and the quality of the execution service (take fee). This simultaneous choice creates an incentive for the owner - say - of two trading platforms like BATS Europe to engage in second-degree price discrimination and set different fee structures across the two trading platforms. [Riccó et al. \(2023\)](#) extend [Chao et al. \(2018\)](#) by considering different regulatory restrictions, a third period and HFTs to show that optimal access pricing depends on the population which characterizes the market, i.e., the amount of ex-ante heterogeneity in investors’ private values, and that with large gains from trade it can result in strictly positive fees. They also show that the widespread use of rebate-based access pricing can be explained by the growing importance of HFT post Reg-NMS.

Among the existing empirical works on maker-taker fees, [Lutat \(2010\)](#) studies the October 2008 introduction of a maker-taker pricing model on the Swiss exchange and finds a decrease in depth but no significant effect on spreads. [Malinova and Park \(2015\)](#) study the 2005 switch by the Toronto Stock Exchange from a value-based to a volume based maker-taker fee schedule that was accompanied by an increase both in the rebate and the take fee, and they find that for the stocks that did not experience a change in total fee, quoted spread declined but cum-fee spreads (quoted spread plus twice the take fee) remained unaffected ostensibly supporting [Colliard and Foucault \(2012\)](#). As [Malinova and Park \(2015\)](#), we consider markets with discrete prices but we test the predictions of our model that unlike [Colliard and Foucault \(2012\)](#) has discrete prices. We find that a decrease in make and take fees is related to changes in both quoted and cum-fee spreads.¹³

[Tham, Sojli, and Skjeltorp \(2018\)](#) using data from the Nasdaq OMX BX and exogenous changes in maker-taker fees show that cross-side liquidity externalities exist and conclude that the reason is that an increase in market makers’ monitoring benefits market takers (“liquidity supply begets liquidity demand”) as predicted by [Foucault et al. \(2013\)](#). [Black \(2022\)](#) uses the Nasdaq February 2015 experiment which lowered its fee structure temporarily on 14 stocks for a four month period and documents that a simultaneous reduction in make and take fees results in lower market efficiency. [Cardella, Hao, and Kalcheva \(2017\)](#) investigate 108 instances of fee changes for U.S. exchanges in 2008-2010 and find that an increase in take

¹³Using Rule 605 data [O’Donoghue \(2022\)](#) finds that changes in the split of trading fees between liquidity suppliers and demanders affect order choice and thereby execution quality.

fees has a larger impact on trading activity than an increase in make fees. [Brauneis, Mestel, Riordan, and Theissen \(2022\)](#) find a similar finding in their crypto-currency markets study. [He, Jarnecic, and Liu \(2015\)](#) study the entry of Chi-X in Europe, Australia, and Japan and find that Chi-X's market share is negatively related to total trading fees and latency, while positively related to liquidity relative to the listing exchanges. [Clapham, Gomber, Lausen, and Panz \(2021\)](#) study the Xetra Liquidity Provider Program at Deutsche Boerse which introduced liquidity rebates and find that the program results in higher liquidity, larger contribution to market-wide liquidity and a higher market share for the venue implementing the rebates, but that market-wide turnover and liquidity do not change. [Anand, Hua, and McCormick \(2016\)](#) study the 2012 introduction of maker-taker pricing in the NYSE Arca options market, and document that execution costs (including fees) for liquidity demanders decline and that the maker-taker pricing encourages market makers to improve quoted prices. [Comerton-Forde, Grégoire, and Zhong \(2019\)](#) and [Lin, Swan, et al. \(2017\)](#) study the effects of the U.S. tick size pilot on venues with different maker-taker (and inverted) pricing models and document that an increase in the tick size results in redistribution of volume towards inverted fee venues.

We make three important contributions to this literature. First, our cross-sectional results, supported by our model, suggest that maker-taker fee changes have asymmetric effects on market quality and market share for small versus large firms. Specifically, as we show for the BXE's simultaneous reductions in both take fees and rebates, the effects are detrimental for large companies while small companies benefit from these fee changes. Second, by focusing on a multi-market design, we quantify the effects of maker-taker pricing reductions not only for the venues that implement the fee changes but also for their direct competitors. Third, we develop a dual market model where investors can freely choose their order submission strategy and order aggressiveness, based on past, current, and future states of the book.

3. Theoretical Background and Empirical Predictions

3.1. Model

In this section we briefly describe our model.¹⁴ Traders arrive sequentially over the trading game that lasts N periods, $t_z = t_1, \dots, t_N$ and in the spirit of [Riccó, Rindi, and Seppi \(2022\)](#) we consider two different specifications with different investor trading activity, one with three periods, $N = 3$, $t_z = \{t_1, t_2, t_3\}$, and one with four periods, $N = 4$, $t_z = \{t_1, t_2, t_3, t_4\}$. At each period t_z a risk-neutral investor arrives at the market with a private valuation equal to γ_{t_z} which is an i.i.d. drawn from a uniform distribution, $\gamma \sim U[\underline{\gamma}, \bar{\gamma}]$, $\underline{\gamma}$ being the lowest valuation and $\bar{\gamma}$ the highest valuation traders may have.¹⁵ The support width $S = \bar{\gamma} - \underline{\gamma}$ -

¹⁴See Appendix 1 for more detailed discussion on the model solution.

¹⁵As in [Colliard and Foucault \(2012\)](#) and in many other models of limit order books (e.g., [Hollifield, Miller, and Sandås \(2004\)](#); [Goettler et al. \(2005\)](#); or [Goettler, Parlour, and Rajan \(2009\)](#)) heterogeneity in investors'

symmetrically distributed around the asset value AV - indicates the dispersion of traders' gains from trade.¹⁶ Traders coming to the market with extreme values of γ_{t_z} are more eager to trade by taking liquidity, whereas traders arriving with γ_{t_z} values close to AV are more willing to supply liquidity. Trade size is unitary.

We model two identical limit order books that we label Market I (mrkI) and Market II (mrkII) respectively. Each limit order book, Market I or Market II, has a grid of four prices, $P_i^j = \{S_2^j, S_1^j, B_1^j, B_2^j\}$, for $j = mrkI, mrkII$, two on the ask and two on the bid side of the book around the same asset value AV . Both trading platforms have a tick size equal to τ , so the ask prices are equal to $S_1^j = AV + \frac{1}{2}\tau$ and to $S_2^j = AV + \frac{3}{2}\tau$ respectively for the inside and outside quotes, and symmetrically the bid prices are equal to $B_1^j = AV - \frac{1}{2}\tau$ and to $B_2^j = AV - \frac{3}{2}\tau$. The state of the limit order book of market j at time t_z is the vector $lob_{t_z}^j = \left\{ l_{t_z}^{P_i^j} \right\}$, where $l_{t_z}^{P_i^j}$ is the depth (number of orders/shares) of the limit order book j at price P_i^j at time t_z .

In each period t_z , a trader arrives, observes the state of the two limit order books and chooses among different possible trading strategies, $y_{t_z}^j$, where Y_{t_z} is the set of possible trading strategies at time t_z . Table 1 reports the payoffs from the different orders that a trader can choose at t_z when arriving either at Market I or at Market II. An investor can choose to submit a limit order ($LO_{t_z}^j(P_i^j)$) or a market order ($MO_{t_z}^j(P_i^{j,b})$) either to Market I or to Market II, or can alternatively decide not to trade (NT_{t_z}).¹⁷ Hence, $Y_{t_z} = \{LO_{t_z}^j(P_i^j), MO_{t_z}^j(P_i^{j,b}), NT_{t_z}\}$.

At t_1 both Market I and Market II open empty and therefore traders will only be able to offer liquidity by posting limit orders. At t_2 (t_2 and t_3) traders can either take or make liquidity via market or limit orders, and at t_3 (t_4), which is the last period of the trading game if $N = 3$ ($N = 4$), traders will only submit market orders or decide not to trade as the execution probability of a limit order is zero. Conditional on their private valuation and the state of the two limit order books, traders opt not to trade (NT_{t_z}) in any period t_z when the payoffs of the possible $LO_{t_z}(P_i^j)$ and $MO_{t_z}(P_i^{j,b})$ are non-positive. Traders face trading fees that can be positive or negative (rebates). In particular a trader will face a take fee TF (tf) if he takes liquidity by posting a market order on Market I (Market II); a trader will face a make fee MF (mf) if he posts a limit order on Market I (Market II). For example, if Market I opts for a maker-taker pricing structure that consists in a positive take fee ($TF > 0$) and a negative make fee ($MF < 0$) a market participant sending a market order to Market I will have to pay a take fee to the trading platform when the market order is executed. Traders opting instead

private valuations generates gains from trade.

¹⁶For simplicity, in our comparative statics we keep the support constant. However, if we solve the model to evaluate the effects of - for example - an increase in the support, we find results on the equilibrium trading strategies and market quality metrics that are consistent with the results we obtain when evaluating the effects of an increase in the number of trading periods.

¹⁷We label the best ask and the best bid prices with the superscript "b".

to post a limit order to Market I will receive a rebate when the limit order is executed. In this case, the rebate is a reward that traders receive when they supply liquidity to the limit order book, whereas the take fee is a charge traders have to pay when they take liquidity.

Consistent with real market practice, both Market I and Market II are governed not only by price but also by time priority rules. If at time t_1 a trader posts, for example, a limit sell order to Market I at the second level of the book, the next period a trader can undercut the resting limit order by posting a more aggressive limit sell order on the first level of either Market I or Market II limit order book. Furthermore, he can hit the limit order initially posted to Market I with a market buy order, or he can post a limit buy order to Market II at the first or at the second level of the book. He can finally decide not to trade. Orders cannot be cancelled or modify. It should be noted that in our model investors decide to trade after observing the state of the limit order book. In addition, when choosing their order type, investors estimate the probability of execution of their limit orders by taking into account all of the future possible states of the book of both Market I and Market II. Investors' optimal order choice is fully endogenous in our model. We do not restrict buyers and sellers to alternate, or to necessarily undercut existing orders, which are standard assumptions used to sustain a stationary equilibrium.¹⁸ Our model does not rely on a stationary equilibrium but with a limited number of periods allows us to obtain a stylized framework of intra-day trading which captures all of the transmission mechanisms in place when a trading venue changes its fee pricing.

Formally, a trader arriving at time t_z will choose the order, $y_{t_z}^j$, that maximizes the expected payoff, $\pi_{t_z}^j$, given his private valuation of the asset, γ_{t_z} , the state of the two limit order books, $lob_{t_z-1}^j = \{l_{t_z-1,i}^{P^j}\}$, and the trading fees, Ω^j , where $\Omega^{mrkI} = \{MF, TF\}$ and $\Omega^{mrkII} = \{mf, tf\}$:

$$\begin{aligned} & \max_{y_{t_z}^j \in Y_{t_z}} \pi_{t_z}^j(y_{t_z}^j \mid \gamma_{t_z}, lob_{t_z}^{mrkI}, lob_{t_z}^{mrkII}, \Omega^{mrkI}, \Omega^{mrkII}, N, S) \\ &= \max_{y_{t_z}^j \in Y_{t_z}} \pi_{t_z}^j(y_{t_z}^j \mid \Lambda_{t_z}) \end{aligned} \quad (1)$$

where $\Lambda_{t_z} = \{\gamma_{t_z}, lob_{t_z}^{mrkI}, lob_{t_z}^{mrkII}, \Omega^{mrkI}, \Omega^{mrkII}, N, S\}$. When choosing their order submission strategies, traders face a trade-off between non-execution costs and price opportunity costs. If they opt for $MO_{t_z}^j(P_i^{j,b})$, they get immediate execution at the best ask price, $S_{t_z}^{j,b} = \min \{S_{t_z,i}^j \mid l_{t_z,i}^{S^{mrkI}}, l_{t_z,i}^{S^{mrkII}}, \Omega^{mrkI}, \Omega^{mrkII}, N, S\}$ if it is a buy order or at the best bid price, $B_{t_z}^{j,b} = \max \{B_{t_z,i}^j \mid l_{t_z,i}^{B^{mrkI}}, l_{t_z,i}^{B^{mrkII}}, \Omega^{mrkI}, \Omega^{mrkII}, N, S\}$, if it is a sell order, where $l_{t_z,i}^{S^j}$ ($l_{t_z,i}^{B^j}$) indicates the number of shares available at the i -th price level of the ask side (bid side)

¹⁸See [Graziani, Rindi, and Yueshen \(2023\)](#) and [Riccó et al. \(2022\)](#) for a discussion on cancellation and the role of stationary equilibrium.

of the j -th market. If instead they choose a $LO_{t_z}^j(P_i^j)$, they face execution uncertainty but they will get a better price if the order executes. When the expected payoffs for an order routed either to Market I or to Market II are the same, we assume that the trader randomizes and routes the order with equal probability to both trading platforms.

Following Colliard and Foucault (2012), the model is solved by backward induction, and as in Chao et al. (2018), conditional on the pricing grid characterized by the “relative tick size”, i.e., the tick size, τ , relative to the support of traders’ valuation, S , it has a closed-form solution for each set of trading fees, Ω . We start from the end of the trading game, t_3 (for $N = 3$), when traders rationally submit only $MO_{t_3}^j(P_i^{j,b})$, and solve the model for the equilibrium market buy and market sell orders, $y_{t_3}^j$. As the equilibrium probabilities of market buy and market sell orders at t_3 are the execution probabilities of $LO_{t_2}^j(P_i^j)$ (to sell and to buy respectively) at t_2 , the model can then be solved at t_2 , and recursively at t_1 (see Appendix 1). Similar arguments hold for $N = 4$. The obtained equilibrium order submission probabilities allow us to draw predictions on order flows and also to build market quality metrics.

Limit orders in each period t_z and in each market j , $LO_{t_z}^j(P_i^j)$, are computed as the weighted average of the probability of observing a limit order conditional on the different equilibrium states of the book, $lob_{t_z}^j$, where the weights are the probabilities of the different states of the book in period t_z : $E[LO_{t_z}^j | \Lambda_{t_z}]$. Market orders, $MO_{t_z}^j(P_i^{j,b})$, are computed in a similar way: $E[MO_{t_z}^j | \Lambda_{t_z}]$. We build measures of quoted spread (*Quoted Spread* ^{j}), depth at the best bid-offer (*BBODepth* ^{j}), and market share (*MS* ^{j}) based on the equilibrium limit orders (LO^j) and market orders (MO^j) submission probabilities. In each period t_z and in each market j the quoted spread, *Quoted Spread* _{t_z} ^{j} , is computed as the weighted average of the probability of observing the inside spread conditional on the different equilibrium states of the book, $lob_{t_z}^j$, the set of fees involved, Ω^j , and the length of the trading game, N , where - as before - the weights are the probabilities of the different states of the book in period t_z : $E\left[\left(S_{t_z,i}^{j,b} - B_{t_z,i}^{j,b}\right) | \Lambda_{t_z}\right]$.¹⁹ Depth at the best bid-offer, *BBODepth* _{t_z} ^{j} is computed as the weighted average of the sum of the shares available at the best bid and ask prices, $E\left[\left(l_{t_z}^{S^{j,b}} + l_{t_z}^{B^{j,b}}\right) | \Lambda_{t_z}\right]$. We then average *Quoted Spread* ^{j} and *BBODepth* _{t_z} ^{j} over all the periods of the trading game: when $N = 3$ we compute the averages over the three periods, t_1 , t_2 and t_3 ; when $N = 4$ we compute the averages over t_1 through t_4 . Market share for Market I, *MS*^{*mrkI*}, is measured as the average of market orders in Market I over the same trading periods, divided by the sum of the average of market orders in Market I and Market II over the same periods, e.g., $MS^{mrkI} = \frac{\sum_{t_z} MO_{t_z}^{mrkI}}{\sum_{t_z} MO_{t_z}^{mrkI} + \sum_{t_z} MO_{t_z}^{mrkII}}$. We measure market share for Market II in a similar

¹⁹In our model liquidity supply is endogenous. However, when computing the quoted spread, we assume that when the book is empty, at either the ask or the bid side, the maximum possible spread is five ticks. We have also derived effective spread which does not rely on this assumption, but to economize space and consistently with our empirical analysis, we only report *Quoted Spread* ^{j} and *BBODepth* ^{j} .

way.²⁰

We use our models to discuss the effects of a change in fees in Market I on the equilibrium order submission probabilities and the derived order flows and market quality metrics of both Market I and Market II. While our framework may be considered a stylized model of intraday trading, all results are averaged across the different periods of the trading game. Therefore our results allow us to draw predictions on how a change in trading fees affects the overall activity of the trading day captured by the daily data we use for our empirical analysis.

3.2. Model Results and Empirical Predictions

In this section we discuss the mechanisms that according to our models drive the change in market quality following adjustments in trading fees. We aim to provide insights for our empirical experiment, where BATS reduced the rebate on liquidity provision for CXE and decreased both the rebate and the take fee for BXE. First, we analyze the effects of a rebate change, followed by an assessment of the combined impact of changes in both the maker fee (MF) and the taker fee (TF).²¹

Results on the effects of a change in rebate on liquidity provision in Market I are presented in Table 2.²² Results on the effects of a change in both the take fee and the rebate on liquidity provision are presented in Table 3.²³ Both tables report results for the two markets with 3-period ($N = 3$, Panel A) and 4-period ($N = 4$, Panel B) setups, respectively. This allows us to investigate how our results change as the number of trading periods increases and the market experiences a higher level of trading activity.

To understand the effects on market quality - while keeping all other factors constant - of an increase in trading activity, consider the results for the equilibrium order submission probabilities of both limit and market orders, as well as the derived metrics of market quality reported for the benchmark market in columns (1) of Table 2 (or of Table 3). These results are obtained by solving the model for the benchmark regime with all of the trading fees set to zero, $\{MF, TF, mf, tf\} = \{.00, .00, .00, .00\}$.

All else equal, when trading activity increases from $N = 3$ to $N = 4$, orders have an additional period to execute, resulting in an increase in both liquidity demand and overall liquidity

²⁰Appendix 1 shows how the metrics of market quality are obtained starting from the equilibrium order submission probabilities.

²¹To streamline our discussion, we present results for fee values that allow us to discuss the predominant effects at play. Our findings hold robust across all parameter values within the considered fee ranges.

²²In real markets trading fees are generally structured such that total fee, which is the sum of MF and TF, is positive. This ensures that the exchange earns a profit upon execution of each order. To isolate the controlled effect of a change in rebate, we analyze the impact of the change in rebate on liquidity provision in Market I while keeping the taker fee (TF) constant at zero. Results are qualitatively the same if we consider a reduction of the rebate on liquidity provision in a market with a maker-taker fee structure, i.e., with $TF > |MF| > 0$, thereby maintaining a positive total fee throughout our comparative static analysis.

²³Results for Market II are reported in Tables A6 and A7 in Appendix 1, respectively.

supply. This explains why both volume (MO) and $BBODepth$ increase and $Quoted Spread$ improves. Tables A6 and A7 in the appendix show that the same results hold for Market II as the model is symmetric.

Taken together, these results show that we can solve our dual market framework under two scenarios. The scenario with higher trading activity (four trading periods) leads to smaller spreads and greater depth, which we use as a proxy for *Large Stocks*. Conversely, lower trading activity (three trading periods) leads to larger spread and lower depth, and we consider this scenario a proxy for *Small Stocks*.²⁴

3.2.1. Change in Make Fee - MF

We start by changing the rebate on liquidity provision only in Market I, holding all other fees constant at zero, $TF, mf, tf = 0.00, 0.00, 0.00$. We isolate the change in the rebate to understand the causal effects that such a change in trading fees determines on the quality of both Market I and Market II when they compete for liquidity provision.²⁵ We solve our models for three sets of trading fees in Market I: $\{MF, TF\} = \{.00, .00\}$, $\{MF, TF\} = \{-.001, .00\}$, and $\{MF, TF\} = \{-.005, .00\}$, while holding the trading fees in Market II constant at zero: $\{mf, tf\} = \{.00, .00\}$.

Throughout our simulations we consider changes in the fee breakdown which may or may not be associated with a change in total fees. When prices are discrete due to the existence of a tick size a change in the fee breakdown matters irrespective of the change in total fees. For this reason, we focus on the relative changes in the make and take fees as opposed to the change in the total fee.²⁶

Table 2 reports results for Market I whereas Table A6 reports results for Market II. We first discuss the effects of a change in rebate in the regime with three trading periods (Panel A, columns 1 through 5), and then show how results change when there are four trading periods (Panel B, columns 1 through 5). This allows us to compare our results for the *Large Stocks* scenario versus the *Small Stocks* scenario.

Panel A columns 1 through 3 show the effects of a change from a regime without fees,

²⁴In Tables 2 and 3 for Market I (and A6 and A7 for Market II in the Appendix) we highlight in gray the results we obtain on our market quality measures: *Volume*, *QuotesSpread*, *BBODepth* and *MS*. We refer to these predictions when empirically testing these variables using data from BATS.

²⁵Although trading platforms may occasionally set their pricing such that the total fee (maker fee plus taker fee) is negative for short periods, they generally set fees to ensure a positive total fee. We change the maker fee to investigate the trade-offs that govern our model. The more realistic case of an increase in the total fee will be discussed in Section 3.2.2.

²⁶Consistent with Foucault et al. (2013), we show that when prices are discrete, a symmetric change in the fee breakdown cannot be neutralized by investors' order submission strategies, leading to a change in the cum-fee spread. With discrete prices, the fee breakdown matters irrespective of the change in the total fee. Therefore, what is relevant is the relative change in the make and take fees rather than the change in the total fee. In fact, it can be shown that different fee breakdowns leading to the same change in total fee can differently affect the quoted spread.

$\{MF, TF\} = \{.00, .00\}$, to a regime with a rebate on liquidity provision, $\{MF, TF\} = \{-.001, .00\}$. Results reported in column 4 show the equilibrium order submission probabilities associated with a further increase in rebate, $\{MF, TF\} = \{-.005, .00\}$, and results in column 5 show the percentage change ($\Delta\%$) in the equilibrium order submission probabilities following the change in regime from $\{MF, TF\} = \{-.001, .00\}$, to $\{MF, TF\} = \{-.005, .00\}$.

The increase in rebate in Market I (Table 2) enhances traders' willingness to supply liquidity resulting in an increase in LO . The increased propensity to offer liquidity intensifies competition for the provision of liquidity, deepening the order book. Consequently, market quality improves with *Quoted Spread* decreasing and *BBODepth* increasing. Moreover, the increase in limit orders stimulates an increase in market orders (MO), thereby boosting trading volume as well.

MS increases substantially in Market I but some activity still survives on Market II (Table A6).²⁷ When the queues on Market I become overly long, investors switch to the top of Market II where they forgo the rebate but gain higher execution probabilities. As TF is zero in both Market I and Market II, liquidity takers are indifferent between taking liquidity from Market I or from Market II.²⁸ The same line of reasoning would not apply if one of the two markets had a cheaper take fee, as in that case liquidity takers would only take liquidity from the cheapest market.

When the rebate is increased to $MF = -0.005$, both liquidity supply and liquidity demand further increase with the higher rebate on liquidity provision. This intensifies competition for the provision of liquidity at the inside quotes of Market I, leading to an improvement in market quality. It's important to note that as the rebate increases further, competition becomes more intense at the inside quotes of Market I. This results in a higher probability that the order book will open with an order already posted at the top of Market I, which increases

²⁷Note that the sum of the average equilibrium probability of limit orders, market orders and no trade adds up to one across Table 2 and A6, as well as Table 3 and A7.

²⁸For example, in the 3-period model with $S = [0, 2]$ and $\{MF, TF, mf, tf\} = \{.00, .00, .00, .00\}$, consider the branches of the trading game that start at t_1 with the ask side of Market I - the bid side being symmetric. At t_1 investors post both $LO^{mrkI}(S_2) = LO^{mrkII}(S_2)$ with probability 0.0082 and $LO^{mrkI}(S_1) = LO^{mrkII}(S_1)$ with probability 0.2418; at t_2 if the book opens with a $LO^{mrkI}(S_2)$, investors post $LO^{mrkI}(S_1)$ and $LO^{mrkII}(S_1)$ with probability 0.2488, $LO^{mrkI}(B_2)$ and $LO^{mrkII}(B_2)$ with probability 0.0123, and $MO^{mrkI}(S_2)$ with probability 0.4779; if instead at t_2 the book opens with $LO^{mrkI}(S_1)$, investors post $LO^{mrkII}(S_1)$ with probability 0.4959, $LO^{mrkI}(B_2)$ and $LO^{mrkII}(B_2)$ with probability 0.0082, and $MO^{mrkI}(S_1)$ with probability 0.4878.

When all else equal the rebate on MF is increased in Market I, with $MF = -0.001$ and $\{TF, mf, tf\} = \{.00, .00, .00\}$, at t_1 investors post limit order only on Market I that now grants a rebate on MF. More precisely they post $LO^{mrkI}(S_2)$ with probability 0.0159 and $LO^{mrkI}(S_1)$ with probability 0.4841. At t_2 if the book opens with a $LO^{mrkI}(S_2)$, investors do not post - as in the case without fees - limit orders on Market II. They post both $LO^{mrkI}(S_1)$ and $LO^{mrkII}(B_2)$ on Market I with a much greater probability, 0.4975 and 0.025 respectively, and post $MO^{mrkI}(S_2)$ with probability 0.4775. If instead at t_2 the book opens with $LO^{mrkI}(S_1)$ investors have to resort to Market II to post aggressive limit orders at the first level of the book and therefore post $LO^{mrkII}(S_1)$ with a high probability 0.4955; they also post $LO^{mrkI}(B_2)$ with probability 0.01712, and $MO^{mrkI}(S_1)$ with probability 0.4873. This explains the increase in limit orders of Market I, but also the reason why a good deal of activity survives at the top of Market II when a rebate is introduced in Market I.

the likelihood that investors will switch to the inside quotes of Market II ($LO(P_1)$ in Table A6), thereby reducing MS on Market I.

Columns 2 through 5 in Table A6 show the effects of increasing the rebate on liquidity provision ($MF = -0.001$) and further enhancing it ($MF = -0.005$) on the equilibrium probabilities of limit and market order submissions, as well as market quality metrics in Market II. As limit and market orders migrate to Market I, both LO and MO decrease substantially in Market II. However, due to increased competition for liquidity provision at the top of both limit order books, a significant proportion of limit orders ($LO(P_1)$) remains in Market II at the first level of the book, sustaining $BBODepth$ and mitigating the negative effect on $Quoted, Spread$. The $Quoted, Spread$ slightly improves when the rebate is further enhanced in Market I. These results are consistent throughout our two protocols.

Trading Activity

Results reported in Table 2 for Market I and in Table A6 for Market II show that the effects of an increase in the rebate (more negative MF) on both liquidity provision (LO) and market quality ($Quoted Spread$ and $BBODepth$) become stronger when the number of trading periods increases, but market share (MS) decreases. In the 4-period framework the driving effect on liquidity supply of Market I (LO) is stronger resulting in stronger effects on market quality, as proxied by $Quoted Spread$ and $BBODepth$. Even though Market I offers a rebate on liquidity provision, in the 4-period model, overall investors post orders to Market II with a higher probability compared, all else equal, to the 3-period framework. The reason is that there is now an additional period in which the Market I book can open with a limit order posted at the inside quotes, in which case investors may find it profitable to post their orders at the inside quotes of Market II. The increased limit orders posted to Market II explain why MS of Market I improves less when a rebate on liquidity provision is introduced in a market with higher trading activity.

Taken together, these findings lead to our Proposition 1.

Proposition 1. *All else equal, consider the introduction of a rebate on liquidity provision in one limit order book - Market I - that competes with an identical limit order book - Market II. Both limit order books can be characterized by either high or low trading activity ($N = 4$ or $N = 3$ trading periods):*

- *Liquidity supply and hence liquidity demand cluster on Market I resulting in an improvement in quoted spread and depth at the best bid-offer.*
- *Liquidity supply and liquidity demand decrease in Market II and migrate to Market I but*

some activity survives in Market II due to competition for the provision of liquidity at the inside quotes.

- *When trading activity increases, competition for the provision of liquidity increases:*
 - *Market quality improves.*
 - *Migration of order flows from Market I to Market II increases resulting in smaller increase of MS in Market I and smaller reduction of MS in Market II.*
- *When the rebate is further increased on Market I, market quality further improves but the increased competition for the provision of liquidity on Market I induces traders to post limit orders on Market II, resulting in a migration of order flows from Market I to Market II and a negative effect on MS of Market I.*

Appendix 1 show how to derive analytically the equilibrium order submission strategies that lead to the results obtained in Proposition 1. Considering that the effects of a change in the rebate on liquidity provision are stronger when trading activity increases, we can draw our empirical predictions by referencing the *Large Stocks* and the *Small Stocks* scenarios. Consistently with our empirical experiment, we consider a reduction - rather than an increase - of a rebate on liquidity provision:

Empirical Prediction 1. *If Market I decreases its rebate on liquidity provision relative to Market II, order flows migrate out of Market I into Market II, causing market quality to deteriorate and market share to decrease in Market I, while market quality improves and market share increases in Market II.*

The positive effects on market quality (Quoted Spread and BBOdepth) for Market II are stronger for large stocks characterized by greater trading activity. However, the positive effect on market share is weaker. Due to the increased competition for the provision of liquidity on Market II there are more opportunities for investors to migrate from Market II to Market I, resulting in a smaller improvement in the market share (MS) of Market II.

3.2.2. Change in Make Fee and Take Fee - MF&TF

We now study the combined effects of changing both the MF (rebate) and the TF on Market I, while holding fees in Market II constant at zero, $\{mf, tf\} = \{.00, .00\}$. Tables 3 and A7 present the results for Market I and Market II, respectively. As with the analysis of the change in MF, we also explore three different sets of trading fees in Market I for changes in both MF and TF: $MF, TF = .00, .00$, $MF, TF = -.001, .001$, and $MF, TF = -.001, .002$."

As before, we begin by examining the effects of increasing both the rebate on limit orders and the taker fee (TF) in Market I using our 3-period model. Table 3, Panel A, columns 1 through 3, illustrate the transition from a regime without fees, $MF, TF = .00, .00$, to a regime with a negative MF and a positive TF, $MF, TF = -.001, .001$. This setup allows us to examine, all else equal, the isolated effects of a change in TF. Column 4 presents the equilibrium order submission probabilities under a more realistic maker-taker regime where the TF is greater in absolute value than the rebate, resulting in an increased total fee, $MF, TF = -.001, .002$ compared to the maker-taker regime in column 2. Column 5 indicates the percentage change ($\Delta\%$) in equilibrium order submission probabilities resulting from the shift from $MF, TF = .00, .00$ to $MF, TF = -.001, .002$. The same columns in Panel B show the results of the effects of the same change in trading fees in our 4-period model.

When a rebate on liquidity provision is introduced in Market I together with a positive TF of the same magnitude, both liquidity supply (LO) and liquidity demand (MO) decrease in Market I. This effect is particularly pronounced for limit and market orders at the inside quotes, leading to a substantial decrease in $LO(P_1)$ and $MO(P_1)$. The result is a migration of order flows from Market I to Market II, paired with a deterioration of all our metrics of market quality for Market I and an improvement of the same metrics for Market II. As discussed in Section 3.2.1, the introduction of only a rebate on liquidity provision in Market I results in an overall migration of order flows to Market I and an improvement in market quality. Comparing this overall positive outcome with the effect of increasing both the rebate on liquidity provision and the TF, we infer that the net effect of increasing the TF reverses the positive impact of the rebate on liquidity provision alone, leading to a decrease in order flows, especially at the first level of the book in Market I. Liquidity suppliers are aware that although they might receive a rebate by placing a limit order on Market I, their order's execution probability would diminish to zero if any limit order at the same price level were available on the cheaper Market II. Therefore, aggressive liquidity suppliers prioritize Market II. Some less aggressive liquidity suppliers opt to post their limit orders at the second price level of Market I. They do so because they anticipate that when liquidity at the top of the Market II book is depleted, liquidity takers arriving sequentially will have to execute against their limit orders. In such a scenario, albeit with a low probability, they would qualify for a rebate. The fee change from $MF, TF = .00, .00$ to $MF, TF = -.001, .002$ reported in Panel A, column 4 and 5 of Table 3 illustrates the effects of a more realistic change in fees, yielding qualitatively similar results. In the 3-period model, the differences are negligible.

Table 3 and Table A7 respectively illustrate how our results evolve when the number of trading periods increases to $N = 4$ in Panel B. With four trading periods instead of three, patient traders have an additional opportunity to utilize the rebate on liquidity provision by posting limit orders at the outside quotes of Market I. This increased trading activity

contributes positively to market quality, thereby offsetting to some extent the negative effects of the positive TF on Market I. Thus, a higher number of trading periods make traders more inclined to capitalize on the rebate offered by Market I, despite the positive charge on TF. These findings lead to our second set of main results:

Proposition 2. *All else equal, consider the simultaneous increase of the rebate on liquidity provision and of the TF in one limit order book - Market I - that competes with an identical limit order book - Market II. Both limit order books can be characterized by either high or low trading activity ($N = 4$ or $N = 3$ trading periods):*

- *Liquidity supply and liquidity demand migrate from Market I to Market II, causing market quality, proxied by quoted spread and depth at the best bid-offer, to deteriorate in Market I*
- *Liquidity supply and hence liquidity demand cluster on Market II, resulting in a general improvement in quoted spread and depth at the best bid-offer in Market II.*
- *The migration of orders flows to Market II and the resulting effects on market quality and market share of both Market I and Market II are weaker when the markets are characterized by higher trading activity, $N = 4$.*

Appendix 1 show how to derive analytically the equilibrium order submission strategies that lead to the results obtained in Proposition 2. Taken together, our results show that when a rebate on liquidity provision coupled with a positive TF is increased in Market I that competes with an identical Market II, order flows migrate to Market II. Consequently, market quality and market share deteriorate on Market I and improve on Market II.

The negative effect on market quality is weaker when the two markets are characterized by higher trading activity. Hence, overall, our *Large Stocks* scenario should be characterized by a weaker negative effect on market quality following the change in the fee structure than our *Small Stocks* scenario, which instead should be characterized by a stronger negative effect on both market quality and market share.

We can therefore draw our Empirical Prediction 2 by summarizing our results for a reduction rather than an increase in the rebate on liquidity provision and TF, still taking into account our two combined scenarios of *Large Stocks* and *Small Stocks*:

Empirical Prediction 2. *If Market I decreases both its rebate on liquidity provision and its take fee relative to Market II, the activity at the inside quotes of Market I increases, resulting in improved market quality and market share in Market I and deteriorated market quality and*

market share in Market II. The improvement in market quality and market share is stronger for small stocks and weaker for large stocks.

Following the reduction in the rebate on liquidity provision and the TF, less active markets will benefit the most from the new fee structure. Investors in these markets will react less aggressively to the rebate reduction and more aggressively to the TF reduction, thereby minimizing the negative effect of the reduced rebate and magnifying the positive effect of the TF reduction.

4. Data Description and Methodology

4.1. Market Structure and Intermarket Competition

We study the January 1, 2013, changes in BATS maker-taker fees. During our sample period, November 2012 - February 2013, BATS operated two European lit venues, BXE and CXE, and each platform featured a continuous order book executing orders based on price, display, and time priority, and both offered very similar maker-taker pricing at the end of 2012. Table 4 illustrates the trading fee schedules in basis points (bps) that apply for LSE listed firms in each BATS venue as of December, 2012. It shows that the take fee was 0.28 bps (0.30 bps) and the rebate was 0.18 bps (0.20 bps) on BXE (CXE).

BXE and CXE primarily faced competition from the transparent MTF Turquoise (TQ) which also operated a continuous order book executing orders based on price, display, and time priority.²⁹ TQ charged takers 0.30 bps and used a value-based rebate ranging from 0.14 bps to 0.28 bps for monthly value traded above €2.5bn.³⁰ They also faced competition from the LSE which operates a transparent, continuous order book, executing orders based on price, display, and time priority. LSE charged trading fees based on the value-traded using a scale ranging from 0.45 bps to 0.20 bps for orders beyond £10bn of value traded.³¹ Value-tiers are typically determined based on monthly value traded, and rebates are distributed and fees collected ex post on a monthly basis.

European stocks were also actively traded on several dark venues during our sample period, including: two venues operated by BATS - BXE-Dark and CXE-Dark - both were dark midpoint order books; a venue operated by the LSE - TQ-Dark - a dark midpoint order book with both continuous and uncross trading which executed orders based on size followed by time priority; and a venue operated by the broker UBS - UBS-MTF - a continuous midpoint

²⁹TQ was originally launched by a consortium of investment banks on August 15, 2008, but was acquired by the LSE on December 21, 2009. See Gresse (2017) for a discussion of the fragmentation of European equity trading.

³⁰For reference, the average December 2012 exchange rate was £0.813/€.

³¹The LSE used maker-taker pricing up to 2009.

order book with price followed by time priority.³² Their fees are summarized in Table 4.

To illustrate the degree of intermarket competition in our sample of stocks, we manually collect daily data from Fidessa (Fragulator) on share volume reported by each venue, and use it to compute the distribution of market shares across our covered venues. Figure 1, (Figure 1a) reports the distribution of market share for November and December, 2012.³³ It shows that LSE trades (continuous and auction) represent 62.5% of share volume, while lit MTFs capture 33.0%, and dark MTFs capture 4.5% of share volume. BATS lit venues' market share is 27.6% and BATS overall market share is 30.2%.

4.2. *Data and Sample*

We rely on a sample consisting of 120 LSE-listed stocks. The sample is constructed using the following stratification methodology. We begin with a sample of all publicly traded companies listed on the LSE that are also traded on either BXE or CXE (using information provided on the BATS website). The reason we screen on existing BATS trading activity is that we cannot measure changes in market quality and market share at the venue-level unless the stock was traded on BATS both before and after the fee change. For these firms we acquire information on daily average market capitalization and daily price for the month of January 2012 using Compustat Global and Bloomberg. This initial sample consists of 355 firms. We then only focus on firms where market capitalization is greater than £500m in order to have sufficient liquidity when we calculate our measures of market quality. From this set of 258 firms, we sample 12 firms (with 6 firms above the median price and six below) within each market capitalization decile and end up with a representative final sample of 120 LSE firms that also traded on BATS.

For each of our sample stock-venue combinations, we calculate our daily market quality measures and market share using Refinitiv DataScope Select cash equities market data. The data includes all intraday best bid and ask prices and associated depth, as well as all trades (price and size) for each covered venue (exchanges and transparent MTFs), time-stamped to the microsecond. We also use Refinitiv end-of-day data to obtain volume, high, low and closing prices.

To capture the effect of BATS fee changes on measure of market quality, we employ a difference-in-difference specification (described in detail in Section 4.4) where we use a similar sample of Australian firms as a control group. We follow the same stratification methodology used for the LSE sample, to choose the 120 firms of this control sample from the population of Australian firms listed in the Australian stock exchange (ASX).

³²BXE Dark, CXE Dark, and TQ Dark all use the midpoint from the LSE market as their reference price.

³³We exclude off-market trades when we calculate market share, which represented 56.5% of share volume for LSE listed firms during November and December 2012.

4.3. Descriptive Statistics

Our model speaks to the effect of a change in maker-taker fees on market quality at the venue level. Therefore, we calculate market quality measures both for the venues that are changing fees, BXE and CXE, and for their main competitor TQ as well as for the LSE. We calculate five different measures of market quality for each venue as follows: volume is the daily number of shares (in 000s) traded using the end-of-day files from Refinitiv Datascope Select; depth is the time-weighted daily average of the intraday quoted BBO depth in shares at the ask-side and the bid-side of each quote; spread is the time-weighted average of the intraday difference between the ask price and the bid price of each quote in units of currency (£); %spread is the time-weighted daily average of the intraday ask price minus the bid price of each quote divided by the midquote (average of the ask and bid prices); and volatility is the difference between the high and low trading price each trading day (using the end-of-day files from Refinitiv Elektron) divided by the high price. Market share is the daily number of shares traded divided by the total number of shares traded across all venues (CXE, BXE, TQ and LSE).

Table 5 reports summary statistics across stocks based on average daily values for each market quality measure at the listing exchange during December 2012. We also report summary statistics for the distribution of market capitalization in millions as well as price levels in British pounds (£). Firms are classified into size terciles based on market capitalization of the firms one year before the first month of the event (i.e., January 2012).³⁴ We report summary statistics for the overall sample (overall) and for the subsamples of the highest (large) and lowest (small) market capitalization terciles. Table 5 shows that the our sampling methodology ensures that we have a significant dispersion in firm characteristics as well as market quality measures. As expected, size and price are higher and market quality better for large than for small firms.

We compare market quality measures at each MTF venue (BXE, CXE, and TQ) to the LSE in Figure 2 for the period December 2012 (which is the pre-period for our event study analysis which is described in the methodology section below). The figure also depicts whether the venue mean is significantly different from the listing exchange mean based on a simple differences in group means test. All differences are statistically significant with the exception of BXE %spread and CXE volatility, in both cases for the large sub-sample. Furthermore Figure 2 demonstrates that the listing exchange is the dominant venue in terms of share of volume and this is true both overall, and for large and small stocks. CXE captures the second largest fraction of share of volume, and its share of average volume is higher for large than for small stocks. By comparison, both TQ and BXE are smaller players in terms of market share.

MTFs on average have lower market quality than the listing exchange (lower depth and

³⁴Similarly, in unreported results we examine sub-samples based on the median price level (low and high priced stocks).

wider spreads). Comparing MTFs, depths relative to the listing exchange are highest for CXE, while they are almost identical for BXE and TQ. MTFs spreads relative to the listing exchange are lowest for CXE and TQ, while BXE has the widest spreads. Finally, volatility is significantly lower on the MTFs compared to the listing exchange overall, and is also more muted on the MTFs for small than for large stocks.

4.4. Methodology

In order to examine whether the fee changes have a significant effect on market quality and market share for BATS and its competitors, we conduct an event study using an event window of two months centered on the fee-change event.³⁵ We face the usual trade-off when selecting the event window. Using a longer time series would enable us to more precisely measure variables pre- and post-event and also capture longer term effects of the pricing changes. However, a narrower window allows us to reduce the potential effects of confounding factors.³⁶

We start by studying time-series of average daily market quality measures. Specifically, we compute equal-weighted daily means across stocks for each venue both for the overall sample (120 firms) and for sub-samples based on size terciles. The result is four time-series (overall, large, medium, and small) of roughly forty daily observations (trading days) for each venue (BXE, CXE, TQ and LSE).³⁷ We evaluate the change in volume (log), quoted depth (log), quoted spread, and market share for each venue and sample following the fee changes based on a time-series regression:

$$y_t^V = \mu + \delta \cdot Post_t + \varepsilon_t \quad (2)$$

where y_t^V is the measure of market quality for venue V and $Post_t$ is a dummy variable that takes on a value of one for days in the post-event period and zero otherwise. Standard errors are computed using the Newey-West correction for autocorrelation with ten lags.

Recall from the model that the fee changes affect traders' order choice and order routing decisions, and this in equilibrium produces market outcomes that we can measure such as venue market share, volume, depth and spreads. In our empirical setting, all orders routed to a particular venue experience the same fee change so we do not have any within-venue variation across stocks in terms of the fees to exploit for the creation of a control sample (e.g., matching stocks on pre-event characteristics). By contrast, we do have variation in terms of fees across venues trading the same stocks - e.g., BATS changes its fees but fees on TQ and the listing

³⁵We exclude the week of Christmas in December, and instead add the last week of November.

³⁶Our results are qualitatively robust for longer windows (four months before and four months after the fee changes), but the statistical significance is, as expected, lower.

³⁷We winsorize extreme values of the dependent variable at the 1% level for the overall sample to reduce the influence of extreme observations. We also exclude option expiration dates, i.e., for the January 2013 fee change we exclude the 21st of December 2012 and the 18th of January 2013.

exchange remain unchanged. It is therefore tempting to use market quality on competing platforms as a control sample. However, our model shows that traders' response to fee changes affects not just their order choice on the venue which changes its fees, but also affects order inflow from, and order outflow to, competing venues. As a result, market quality on competing venues are likely to be indirectly affected by the BATS fee changes which suggests that we need to investigate both a direct and an indirect effect of the fee changes (Boehmer et al. (2020)).

Therefore, to establish the causal effect of fee changes on measures of market quality, and to address exogenous market trends, we employ a difference-in-difference methodology using Australian firms' market quality measures as a control (control venue ASX). The Australian market is similar to the European market, both in terms of the degree of fragmentation and HFT activity.³⁸ Moreover, there are no trading fee changes in either one of our event windows for the venues trading Australian stocks, making this a good control group. We rely on a sample of Australian stocks that is stratified based on market capitalization and price.³⁹ Specifically, we estimate the following panel regression specification:

$$y_{i,t}^{V,ASX} = \mu + \beta_1 \cdot Treatment_i^V + \beta_2 \cdot Post_t + \beta_3 \cdot Treatment_i^V * Post_t + \eta_{i,t} \quad (3)$$

where $y_{i,t}^{V,ASX}$ is the measure of market quality (either market share, volume (log), spread, or quoted depth (log)) for stocks in venue V and the control venue ASX, subscript i indicates an individual stock, subscript t denotes time in days, $Treatment_i^V$ is a dummy variable that takes on a value of one for stocks in the venue V and zero for stocks in the control venue ASX, and $Post_t$ is a dummy variable that takes on a value of one for days in the post-event period and zero otherwise. Standard errors are clustered by firm and date. The estimated coefficient $\hat{\beta}_3$ measures the change in market quality in venue V associated with the change in trading fees over and above time series changes in market quality that are unrelated to fee changes (captured by the estimated coefficient $\hat{\beta}_2$ of the control venue ASX) and the cross-sectional differences between the market quality measures across venues V and ASX (in the period before the fee changes) captured by the estimated coefficient $\hat{\beta}_1$ of the venue V .

We also estimate a similar difference-in-difference panel version of the relationship between market quality in venue V and the fee changes controlling for market quality on the listing exchange LSE, to evaluate the robustness of our results. Specifically, we estimate the following regression specification:

³⁸HFT activity for European markets for 2013 and 2014 are roughly 25% according to TABB Group, and the level of HFT trading is reasonably steady at 27% of total turnover according to the Australian Securities and Investments Commission report (2015).

³⁹Descriptive statistics of the ASX sample are shown in Appendix 2. They are based on average daily values of each market quality measure at the ASX during December 2012, similarly to Table 5 for the LSE sample. We also report summary statistics of market capitalization in millions as well as price levels, both measured in Australian Dollars (AUD) for the same period as for the LSE sample. For reference, the exchange rate was AUD 1.5/£1

$$y_{i,t}^{V,LSE} = \mu + \beta_1 \cdot Treatment_i^V + \beta_2 \cdot Post_t + \beta_3 \cdot Treatment_i^V * Post_t + \eta_{i,t} \quad (4)$$

where $y_{i,t}^{V,LSE}$ is the measure of market quality for stocks in venue V and the control venue is now the listing exchange LSE. For this analysis, we also use standard errors that are clustered by firm and date. We acknowledge that LSE cannot be used as proper control since it can also be affected by the fee changes –following our theory we expect such indirect effects since we are in an environment with significant intermarket competition. Nevertheless, and following [Boehmer et al. \(2020\)](#) we believe that we can learn from capturing these indirect effects. In particular, whereas in Equation (4), the estimated coefficient $\hat{\beta}_3$ measures the direct change in market quality in venue V associated with the change in trading fees, we note that the estimated coefficient $\hat{\beta}_2$ absorbs any indirect effect caused by spillover from the listing exchange’s response to the fee changes in venue V . Hence, we focus on the joint direct and indirect effect of fee changes ($\hat{\beta}_3 + \hat{\beta}_2$).

5. Empirical Results

In this section, we start by discussing the mapping between theory and the BATS fee changes. We then report results based on the time-series event-study methodology for each Multilateral Trading Facility (MTF): BXE, CXE, and TQ, and for the listing exchange LSE. Next, we report the results based on our main specification: using difference-in-difference panel regressions with a control group of Australian firms to properly control for exogenous market trends. To confirm that our panel regression results are robust, we report similar results based on a number of tests including (1) difference-in-difference panel regressions using trading on the LSE as a control, (2) investigating the fee-changes effects on (a) cum-fee spreads, and (b) a sub-sample of our firms that have matched relative tick sizes.

5.1. Fee Changes and Mapping with the Theory

In late 2012, BATS announced a plan to change its pricing, effective January 1, 2013, of its two transparent trading venues. Specifically, as reported in the second sets of columns in [Table 4](#), BATS eliminated the liquidity rebate from its BXE venue completely (from 0.18 bps to zero) and reduced the take fee from 0.28 bps to 0.15 bps. Furthermore, BATS reduced the CXE liquidity rebate from 0.20 bps to 0.15 bps while leaving the take fee at 0.30 bps. As TQ did not change its pricing, the relative changes in fees/competitiveness across the three European trading venues are the following:⁴⁰

⁴⁰LSE has to a large extent a captive order flow and it offers a flat fee (£0.10) for all order types. As shown in [Table 4](#), it is also substantially more expensive in terms of take fee; hence, we do not expect its market share and market quality to be significantly affected by the change in take fee either.

↓ **rebate MF** : CXE reduced the rebate (negative MF) ($\Delta MF = 5bps$) compared to TQ;

↓ **rebate MF & TF** : BXE reduced the rebate (negative MF) and the take fee (positive TF) ($\Delta MF = 18bps$ and $\Delta TF = -13bps$) compared to TQ;

↓ **rebate MF & TF** : BXE reduced the rebate (negative MF) and the take fee (positive TF) moderately ($\Delta MF = 13bps$ and $\Delta TF = -13bps$) compared to CXE.

Considering all the relative changes in fees across the three MTFs (BXE, CXE, and TQ), according to our model, the net effects of the overall BATS change in pricing can be summarized as:

BXE: An improvement in market quality and market share, stronger for small stocks, as BXE experienced a reduction in both the rebate and the take fee compared to CXE and TQ;

CXE: A deterioration of market quality, stronger for larger stocks, and a reduction in market share, more pronounced for small stocks compared to TQ, resulted from the reduction in rebate only. Moreover, a further deterioration in market quality and market share, stronger for small firms compared to BXE, ensued from BXE's reduction in both the rebate and the take fee;

TQ: An improvement in market quality and an increase in market share for large stocks, along with a deterioration in market quality and a reduction in market share for small stocks. This is the net effect of the improvement in market quality (stronger for large stocks) and market share (stronger for small stocks) due to the reduction in CXE rebate, and of the reduction in market quality and market share (stronger for small stocks) resulting from the BXE reduction in both the rebate and the take fee.

5.2. Collapsed Time-Series Regressions

We first evaluate the effect of BATS' fee changes on volume (log), quoted spreads, quoted depth (log), and market share for each venue for the overall sample and for the two sub-samples based on a collapsed time-series regression following Equation (2). The results in Table 6 show that for stocks overall, volume increase for all venues, but market shares increase only for BXE and TQ, is unchanged for CXE, and declines for the listing exchange. Spreads for stocks overall decline somewhat on BXE and increase on LSE, while we find that depth declines on BXE and increases on TQ.

The results for stocks overall mask significant cross-sectional differences. Consider first small stocks where volume increased only on BXE and LSE. BXE market share of trading small stocks increases at the expense of CXE and TQ. Small stock spreads decline on BXE

without a reduction in depth, but spreads and depths are unchanged on the other MTFs and depth increases on LSE. For large stocks, volume increases on both BATS venues - BXE and CXE, and on TQ. While CXE large stock market share does not change, both BXE and TQ gain market share at the expense of LSE. However, TQ is clearly the winner in terms of gaining market share for large stocks despite not having changed its fees at all. Large stock spreads increase for both BATS markets and depth decreases for BXE. For TQ, depth for large stocks increases with no change in spreads. These results suggest significant cross-sectional difference in the effects of BATS fee changes between large and small firms across venues. In particular, we can detect beneficial market quality effects in TQ market for large firms whereas BXE shows improvements for small firms. In the following subsection, we investigate whether these preliminary conclusions hold up in panel regressions.

5.3. Panel Regressions

We next analyze the effects of the BATS fee changes in a difference-in-difference panel regression specification with Australian stocks as controls, following Equation (3). Note that in this case, the market share regressions compare each venue’s market share of trading LSE-listed stock, e.g. $BXE/(BXE+CXE+TQ+LSE)$, to the market share of ASX of trading ASX-listed stocks, $ASX/(ASX+Australian\ Chi-X)$. Recall that in this specification, we are interested in the interaction coefficient on $Post * Treatment$.⁴¹

Table 7, Panel A reports the results for BXE. Overall, we find that volume increases, spreads decline, and market share increases significantly following the BXE fee changes. By contrast, there is no effect on depth. The results for stocks overall are consistent with our model Prediction 2 that the rebate and take fee reductions encourage order flow to migrate to BXE and market quality improves. The results for sub-samples of large and small stocks also support our model Prediction 2. Small stocks - proxied in the model by stocks populated by investors with lower propensity for liquidity provision and by lower trading activity - benefit the most from the BXE fee reduction and attract both limit and marketable orders with the result of improving market quality and market share. As our model predicts, we expect order flow to come either from CXE or TQ, or from both markets. (Looking at panels B and C we indeed find evidence of that). Large stocks instead do not benefit from the reduction in the fees and experience a deterioration of spreads. This is also consistent with the model Prediction 2 that in a stylized way predicts a much smaller improvement of large stocks’ market quality on BXE since the reduction in the rebate reduces the incentive for liquidity suppliers to build liquidity - at prices beyond the inside quotes.

⁴¹For completeness, we also report in Table 7 the joint effect of $Post+Post * Treatment$ (following [Boehmer et al. \(2020\)](#)), but we focus on that only when we run a difference-in-difference panel regression specification using LSE as a control in the robustness Section 5.4.

Results reported in Panel B for CXE show that, after we control for market developments on our control market, ASX, spreads increase significantly both for the overall sample and for the sub-samples of stocks. In addition market share deteriorates for small stocks (shown by the negative and significant post-variable and the insignificant interaction term). Consistent with our model Prediction 1, these negative effects could be the result of the CXE rebate reduction compared to TQ (i.e., a stronger effect on market quality for large stocks and a stronger effect on market share for small stocks). It could also be the result of the BXE simultaneous rebate and take fee reductions (with a stronger effect on small stocks) consistent with our model Prediction 2. To uncover whether both effects are at play we look at the BATS fee change effects on the competitor market TQ, which did not implement any maker-taker pricing changes.

The results for TQ - reported in Table 7 Panel C - show that for the overall sample, depth and market share increase significantly without any significant change in spreads or volume. For large stocks we find a significant decrease in spreads, a significant increase in depth and also in market share. This significant improvement in market quality and market share is the outcome of the CXE reduction in rebate and the resulting migration of order flow from CXE (consistent with Prediction 1). Note, however, that we do not find any improvements for small firms. In particular, we observe a reduction in market share (shown by the negative and significant post-variable and the insignificant interaction term) suggesting an outflow of orders from TQ to BXE driven by the BXE reductions in rebate and take fee. This asymmetry in the effects on TQ between large and small firms is predicted by our model since the effects of BXE fee changes are expected to be stronger for small firms. Overall, this suggests that both our model Prediction 1 - applied by CXE reduction in rebate - and Prediction 2 - applied by BXE reductions in both rebate and take fee - are necessary to explain the documented market quality and market share changes in TQ large and small firms.

Taken together our results show that the reduction in BATS fees only benefited small firms in the BXE market. The negative externality of BATS strategic change in pricing resulted in an improvement of TQ market quality and market share for large stocks. A market that did not change its fees.

The final panel in Table 7 - Panel D - reports the effect of BATS' fee changes on LSE. We find a significant increase in market share for small stocks together with a marginally significant increase in spreads but no significant changes on volume or depth. For large stocks, we find a marginally significant decrease in market share. Note that the results for LSE are weaker than the results discussed in the previous subsection. The reason is that many of the patterns we discovered for LSE in Table 6 are no different from what we observe in the control sample from ASX, and the panel difference-in-difference regressions corrects for exactly this.⁴²

⁴²LSE market share for small stocks increases relative to ASX because of a secular decline in ASX market

5.4. Robustness

We run one more panel difference-in-difference regression specification in order to verify the robustness of our results. Specifically, instead of using a sample of ASX-listed stocks as controls we use trading of the same stocks on the LSE as a control for trading on BXE, CXE, and TQ, following Equation (4). The results are reported in Table 8 for each of the lit venues that compete with the LSE: BXE, CXE, and TQ. In this specification, the coefficient on *Post* captures the effect on LSE trading of BATS’ fee changes, while the coefficient on the interaction term *Post * Treatment* captures the differential effect on the three venues BXE, CXE, and TQ respectively relative to LSE. Virtually no coefficient on *Post* is statistically significant for volume, spreads, and depth. This is consistent with the results in Table 7, Panel D, which showed that there were no significant changes in volume or market quality for LSE. However, as expected since we have already documented large shifts in LSE market share following BATS’ fee changes, the coefficient on *Post* is highly significant for the market share results in the last three columns. Recognizing the effect of BATS fee changes on LSE, and following [Boehmer et al. \(2020\)](#), we report the sum of the direct and the indirect treatment effects at the bottom of each panel.

Starting with the market share results, LSE market share as captured by the coefficient on *Post* falls by roughly 2.2 percentage points for large stocks and about 1.4 percentage points overall with no change for small stocks. Adding these indirect effects to the interaction coefficient, the results for BXE in Panel A show that the total effect on market share for large stocks is insignificant while market share for small stocks and stocks overall increase of 1.5 and 0.9 percentage points respectively. A similar calculation shows that the total effect on market shares for CXE is a significant increase for large stocks of 0.5 percentage points and a decline for small stocks of 0.9 percentage points, but does not affect overall sample market share on this venue. Similarly, for TQ we find a significant total effect on market share for large stocks and overall of 1.7 and 0.8 percentage points respectively, and a marginally significant reduction in market share for small stocks of 0.4 percentage points. The magnitude and significance of the shifts in market share are similar to those we observed in Table 6. Using LSE as control results provide similar support to our model as Section 5.3.

We also investigate the effect of the fee change on cum-fee spreads (quoted spread plus twice the take fee), following [Malinova and Park \(2015\)](#). We run univariate (time-series) regressions, as shown in Equation (2), for each of our trading venues (BXE, CXE, TQ and LSE). Since the listing exchange (LSE) follows a take fee schedule, we calculate cum-fee spreads for this market based on both the lower (e.g., 0.20 bps for LSE) and upper (e.g., 0.45 bps for LSE) take fees. In contrast to [Malinova and Park \(2015\)](#)—who base their model on [Colliard and Foucault \(2012\)](#) without a tick size—but in support of our model, we find that cum-fee spreads share in this market segment over our sample period.

are affected by fee changes. In particular, our cum-fee results show: (1) an overall increase in CXE cum-fee spreads driven by large firms, and (2) a decrease (increase) in BXE cum-fee spreads in small (large) firms. These results are similar to our quoted spread time-series results in Section 5.2, though the economic significance appears to be smaller.⁴³

Lastly, we address the concern that our finding of a differential effect of fee-changes for large and small firms may be due to differences in relative tick size. We note that in our sample of LSE-listed firms, this is not a relevant factor, because market operating rules force trading prices to be positively correlated with tick sizes (the dynamic tick size rule), thereby ensuring that relative tick size does not vary significantly.⁴⁴ Nevertheless, to be thorough, we do identify and exclude from our sample 7 large firms that follow a slightly different dynamic tick size rule (the SET0 group) for which there are minor differences in tick size. For the remaining sample the relative tick sizes among large and small size groups are identical. We rerun our analysis and confirm that our key finding of a differential effect of fee-changes across large and small firms still holds as the results are quantitatively and qualitatively similar to our main findings.

6. Conclusions and Policy Implications

Maker-taker pricing is actively debated among academics, practitioners, market operators, and is currently under review by U.S. and European regulators. We shed light on this debate by studying how reduced rebates and take fees influence both market quality and market share in a context of inter-market competition.

We first develop a theoretical model with two identical venues, operating limit order books governed not only by price but also by time priority. Our model shows that understanding the flow of orders between venues is crucial for predicting the impact on a venue’s market quality and market share when it alters its maker-taker pricing structure. We then empirically investigate the impacts of changes in maker-taker fees, implemented by BATS on its two European venues - BXE and CXE - in 2013, on market quality and market shares. We compare these outcomes with the predictions from our theoretical model. The model underscores that fee changes are likely to affect competing venues as well. Therefore, we analyze the impacts on market shares and market quality not only on BXE and CXE but also on the competing venue TQ, along with the listing exchange.

BXE eliminated its rebates entirely, and significantly reduced the take-fee. Consistent with our model’s predictions, these fee reductions attracted order flow to BXE, thereby improving market quality and market share. The reduction in the take fee made BXE cheaper, further attracting order flows. While the elimination of rebates had a relatively weaker negative

⁴³Due to space considerations, these results are not reported but are available from the authors upon request.

⁴⁴This is why we do not incorporate relative tick size in our theoretical model.

impact on small stocks, the overall positive effect of BXE’s fee structure change was stronger for small capitalization stocks. CXE reduced only its rebate, and in line with our model predictions, this reduction had a detrimental effect on market quality, resulting in increased spreads. Importantly, to emphasize the impact of intermarket competition on other venues, we also examine TQ, which did not alter any of its fees. We find that market quality and market share significantly improve for large stocks on TQ following BATS’ fee changes.

Based on our empirical results, we conclude that the effects on market quality and the distribution of volume of a proposal such as the one put forth by ICE and SIFMA are likely to differ across stocks. Specifically, our evidence suggests that an elimination of the make fee and a reduced take fee cap would result in worse market quality for large capitalization stocks but better market quality for small capitalization stocks. This suggests that the elimination of make-fees are going to be particularly detrimental for liquid stocks. In light of our findings, BATS’ proposal to eliminate rebates and reduce take fees for the most liquid stocks, while allowing higher rebates and take fees for less liquid stocks, may be ill advised.

Documenting cross-sectional differences in the effect of fee changes in BXE and CXE venues leads naturally to the following question: was the BATS fee change successful? This is a challenging question to answer, as we are unable to observe the counterfactual, i.e., what would have happened had BATS not changed its fees. We believe, nevertheless that we can shed light on this question by examining what happened to (a) market share across venues and (b) (estimated) changes in fee revenues due to the now higher fees. Regarding the former, Figure 1 shows that BATS combined market share in LSE-listed firms declined from 30.2% in November and December 2012 (Figure 1a) to 28.3% in February and March 2013 (Figure 1b). The distribution across BATS venues also shows that the loss of market share was primarily caused by traders leaving CXE which is where rebates were reduced. By contrast, the BXE venue, where both fees were reduced actually gained market share, suggesting that there is a place for a venue without liquidity rebates and low take fees. Regarding the effect on BATS revenues, our initial investigation shows that BATS’ fee changes did lead trading revenues to rise significantly in both venues. In particular, we conservatively estimate a revenue increase of 1.20% for BXE and 2.54% for CXE. Hence, our results suggest that the BATS fee changes were indeed successful in this regard.

We close by highlighting our contributions to the literature. We take intermarket competition between two limit order books into account in both our theoretical and empirical analyses of maker-taker fee changes. Given the significant fragmentation of today’s equity markets, this is clearly an important consideration. We show empirically that the spillover effects on competing venues are significant. Our evidence is corroborated by recent fee experiments conducted by both the Nasdaq and the TSX which lost market share after reducing liquidity rebates.

We also focus our analysis on a multi-platform reduction in rebates and take fee. The previous literature has mainly studied the elimination of a charge for liquidity provision (Lutat (2010)) and increases in the make and take fees (Malinova and Park (2015)). The current policy debate is focused on reducing rather than increasing make-take fees, and our evidence is therefore directly relevant to the proposed changes. We believe that since the SEC Transactions Fee Pilot proposal is currently frozen, our study provides the closest substitute, since the two venues fee-changes closely match the two test groups (only rebate reductions versus simultaneous reductions in make-take fees) proposed by the pilot. Overall, our findings can be used as a guide for how make-take pricing fees affect market quality and market share in multi-market trading environments.

References

- Anand, A., Hua, J., McCormick, T., 2016. Make-take structure and market quality: Evidence from the US options markets. *Management Science* 62, 3271–3290.
- Angel, J. J., Harris, L. E., Spatt, C. S., 2015. Equity trading in the 21st century: An update. *Quarterly Journal of Finance* 5, 1550002.
- Battalio, R., Corwin, S. A., Jennings, R., 2016. Can brokers have it all? on the relation between make-take fees and limit order execution quality. *Journal of Finance* 71, 2193–2238.
- Black, J. R., 2022. The impact of make-take fees on market efficiency. *Review of Quantitative Finance and Accounting* 58, 1015–1035.
- Boehmer, E., Jones, C. M., Zhang, X., 2020. Potential pilot problems: Treatment spillovers in financial regulatory experiments. *Journal of Financial Economics* 135, 68–87.
- Brauneis, A., Mestel, R., Riordan, R., Theissen, E., 2022. The anatomy of a fee change—evidence from the cryptocurrency markets. *Journal of Empirical Finance* 67, 152–167.
- Brolley, M., Malinova, K., 2013. Informed trading and maker-taker fees in a low-latency limit order market, working paper University of Toronto.
- Buti, S., Rindi, B., Werner, I. M., 2017. Dark pool trading strategies, market quality and welfare. *Journal of Financial Economics* 124, 244–265.
- Cardella, L., Hao, J., Kalcheva, I., 2017. Make and take fees in the US equity market, working paper Rawls College of Business.
- Chao, Y., Yao, C., Ye, M., 2018. Why discrete price fragments us stock exchanges and disperses their fee structures. *Review of Financial Studies* 32, 1068–1101.
- Clapham, B., Gomber, P., Lausen, J., Panz, S., 2021. Liquidity provider incentives in fragmented securities markets. *Journal of Empirical Finance* 60, 16–38.

- Colliard, J.-E., Foucault, T., 2012. Trading fees and efficiency in limit order markets. *Review of Financial Studies* 25, 3389–3421.
- Comerton-Forde, C., Grégoire, V., Zhong, Z., 2019. Inverted fee structures, tick size, and market quality. *Journal of Financial Economics* 161, 141–164.
- Foucault, T., 2012. Pricing liquidity in electronic markets. United Kingdom Government Office of Science, Foresight Projects .
- Foucault, T., Kadan, O., Kandel, E., 2013. Liquidity cycles and make/take fees in electronic markets. *Journal of Finance* 68, 299–341.
- Goettler, R. L., Parlour, C. A., Rajan, U., 2005. Equilibrium in a dynamic limit order market. *Journal of Finance* 60, 2149–2192.
- Goettler, R. L., Parlour, C. A., Rajan, U., 2009. Informed traders and limit order markets. *Journal of Financial Economics* 93, 67–87.
- Graziani, G., Rindi, B., Yueshen, B., 2023. Optimal tick size. Working Paper, Università Bocconi .
- Gresse, C., 2017. Effects of lit and dark market fragmentation on liquidity. *Journal of Financial Markets* 35, 1–20.
- Harris, L., 2013. Maker-taker pricing effects on market quotations, working Paper USC Marshall School of Business.
- He, P. W., Jarnecic, E., Liu, Y., 2015. The determinants of alternative trading venue market share: Global evidence from the introduction of Chi-X. *Journal of Financial Markets* 22, 27–49.
- Hendershott, T., Riordan, R., 2013. Algorithmic trading and the market for liquidity. *Journal of Financial and Quantitative Analysis* 48, 1001–1024.
- Hollifield, B., Miller, R. A., Sandås, P., 2004. Empirical analysis of limit order markets. *The Review of Economic Studies* 71, 1027–1063.
- Lin, Y., Swan, P. L., et al., 2017. Why Maker-Taker Fees Improve Exchange Quality: Theory and Natural Experimental Evidence, working Paper University of New South Wales.
- Lutat, M., 2010. The effect of maker-taker pricing on market liquidity in electronic trading systems—empirical evidence from European equity trading, working Paper Goethe University.
- Malinova, K., Park, A., 2015. Subsidizing liquidity: The impact of make/take fees on market quality. *Journal of Finance* 70, 509–536.
- Malinova, K., Park, A., Riordan, R., 2018. Do retail investors suffer from high frequency traders?, working Paper University of Toronto.
- O’Donoghue, S. M., 2022. Transaction fees: Impact on institutional order types, commissions, and execution quality. *Journal of Financial Markets* 60, 100717.

- OICV-IOSCO, 2013. Trading fee models and their impact on trading behaviour. Tech. rep., The Board of the International Organization of Securities Commissions, FR 12/13.
- Riccó, R., Rindi, B., Seppi, D. J., 2022. Information, liquidity, and dynamic limit order markets. Liquidity, and Dynamic Limit Order Markets (February 24, 2022) .
- Riccó, R., Rindi, B., Seppi, D. J., 2023. Optimal fee pricing. Available at SSRN 3779195 .
- Tham, W. W., Sojli, E., Skjeltorp, J. A., 2018. Cross-sided liquidity externalities. Management Science 64, 2901–2929.

Table 1: **Order Submission Strategies**

In this table column 1 reports the payoffs of traders' order submission strategies, $\pi_{t_z}^j$. Column 2 reports the payoff of orders posted on Market I $\pi_{t_z}^{mrkI}$ and column 3 the payoffs of orders posted on Market II $\pi_{t_z}^{mrkII}$ where $j = mrkI$ for Market I and $j = mrkII$ for Market II. Market orders to sell ($MO_{t_z}^j(B_i^{j,b})$) and market orders to buy ($MO_{t_z}^j(S_i^{j,b})$) execute against the best bid price, $B_{t_z}^{j,b} = \max\{B_{t_z,i}^j | l_{t_z,i}^{B^{mrkI}}, l_{t_z,i}^{B^{mrkII}}, \Omega^{mrkI}, \Omega^{mrkII}, N, S\}$, or the best ask price, $S_{t_z,i}^{j,b} = \min\{S_{t_z,i}^j | l_{t_z,i}^{B^{mrkI}}, l_{t_z,i}^{B^{mrkII}}, \Omega^{mrkI}, \Omega^{mrkII}, N, S\}$ respectively, where $l_{t_z,i}^{B^j}$ ($l_{t_z,i}^{S^j}$) is the number of shares available at the i -th price level of the bid side (ask side) of the j -th market at time t_z . Traders have a personal evaluation of the asset which is, $\gamma \sim U[\underline{\gamma}, \bar{\gamma}]$. MF (mf) is the make fee and TF (tf) is the take fee for Market I (Market II). Limit orders to sell, $LO_{t_z}^j(S_i^j)$, and limit orders to buy, $LO_{t_z}^j(B_i^j)$, posted at time t_1, t_2 or t_3 may execute at the limit price, S_i^j and B_i^j , respectively. Limit buy and sell orders execution probabilities are respectively $Pr_{t_z}(S_i^j | lob_{t_z}^{mrkI}, lob_{t_z}^{mrkII}, \Omega^{mrkI}, \Omega^{mrkII}, N, S)$ and $Pr_{t_z}(B_i^j | lob_{t_z}^{mrkI}, lob_{t_z}^{mrkII}, \Omega^{mrkI}, \Omega^{mrkII}, N, S)$; The payoff of no-trade, NT_{t_z} , is 0 in both markets.

Strategy	Payoffs: Market I (mrkI) $\pi_{t_z}^{mrkI}$	Payoffs: Market II (mrkII) $\pi_{t_z}^{mrkII}$
Market Order to Sell: $MO_{t_z}^j(B_i^{j,b})$	$B_i^{mrkI,b} - \gamma_{t_z} - TF$	$B_i^{mrkII,b} - \gamma_{t_z} - tf$
Limit Order to Sell: $LO_{t_z}^j(S_i^j)$	$(S_i^{mrkI} - \gamma_{t_z} - MF) \times$ $Pr_{t_z}(S_i^{mrkI} lob_{t_z}^{mrkI}, lob_{t_z}^{mrkII}, \Omega^{mrkI}, \Omega^{mrkII}, N, S)$	$(S_i^{mrkII} - \gamma_{t_z} - mf) \times$ $Pr_{t_z}(S_i^{mrkII} lob_{t_z}^{mrkI}, lob_{t_z}^{mrkII}, \Omega^{mrkI}, \Omega^{mrkII}, N, S)$
No Trade: NT_{t_z}	0	0
Limit Order to Buy: $LO_{t_z}^j(B_i^j)$ Limit Order to Buy: $LO_{t_z}^j(B_i^j)$	$(\gamma_{t_z} - B_i^{mrkI} - MF) \times$ $Pr_{t_z}(B_i^{mrkI} lob_{t_z}^{mrkI}, lob_{t_z}^{mrkII}, \Omega^{mrkI}, \Omega^{mrkII}, N, S)$	$(\gamma_{t_z} - B_i^{mrkII} - mf) \times$ $Pr_{t_z}(B_i^{mrkII} lob_{t_z}^{mrkI}, lob_{t_z}^{mrkII}, \Omega^{mrkI}, \Omega^{mrkII}, N, S)$
Market Order to Buy: $MO_{t_z}^j(S_i^{j,b})$	$\gamma_{t_z} - S_i^{mrkI,b} - TF$	$\gamma_{t_z} - S_i^{mrkII,b} - tf$

Table 2: **Equilibrium Order Submission Strategies and Market Quality in Market I. Change in MF of Market I only (for Market II $tf=mf=0.000$). 3-period vs. 4-period Model.**

This table reports for Market I the average equilibrium probabilities of the following order flows and market quality metrics: limit orders, $LO(P_i)$, and market orders, $MO(P_i)$, with the orders breakdown for the outside (P_2) and inside (P_1) price levels - for one side of the market, the other side being symmetric, Quoted Spread (*Quoted Spread*), Best Bid and Offer depth (*BBODepth*), and market share (*MS*). The table reports results obtained under two protocols (models): with 3 periods (Panel A) and 4 periods (Panel B). TF and MF for Market I are reported in rows 1 and 2 (for Market II $\{mf, tf\} = \{.000, .000\}$). Results are reported for different values of MF, holding $TF = 0.000$, specifically: for $MF = 0.000$ (column 1), for $MF = -0.001$ (column 2), and for $MF = -0.005$ (column 4). Column 3 reports the percentage change ($\Delta\%$) in the market quality metrics between $MF = 0.000$ and $MF = -0.001$ and column 5 reports the percentage change ($\Delta\%$) in the market quality metrics between $MF = -0.001$ and $MF = -0.005$. The metrics are reported as averages across all periods of the trading game for each of the two protocols (3 and 4 periods, respectively). For both protocols, the asset value $AV = 1$ and tick size $\tau = 0.01$.

Market I TF Market I MF	Market I				
	0.000 0.000 (1)	0.000 -0.001 (2)	$\Delta\%$ (2) and (1) (3)	0.000 -0.005 (4)	$\Delta\%$ (4) and (2) (5)
Panel A. 3-period model					
<i>LO</i>	0.2521	0.3444	36.63%	0.3448	0.11%
<i>LO(P₂)</i>	0.0083	0.0164	99.01%	0.0161	-1.98%
<i>LO(P₁)</i>	0.2438	0.3280	34.52%	0.3287	0.21%
<i>MO (Volume)</i>	0.1251	0.2105	68.25%	0.2106	0.07%
<i>MO(P₂)</i>	0.0041	0.0081	98.96%	0.0079	-2.06%
<i>MO(P₁)</i>	0.1211	0.2024	67.22%	0.2027	0.15%
<i>No Trade</i>	0.2456	0.2454	-0.09%	0.2446	-0.34%
<i>Quoted Spread</i>	0.0409	0.0374	-8.53%	0.0373	-0.13%
<i>BBODepth</i>	0.4624	0.6414	38.70%	0.6437	0.36%
<i>MS</i>	0.5000	0.8411	68.21%	0.8409	-0.01%
Panel B. 4-period model					
<i>LO</i>	0.2526	0.3836	51.87%	0.3839	0.08%
<i>LO(P₂)</i>	0.0682	0.1362	99.64%	0.1354	-0.53%
<i>LO(P₁)</i>	0.1844	0.2474	34.20%	0.2485	0.42%
<i>MO (Volume)</i>	0.1556	0.2521	61.95%	0.2523	0.08%
<i>MO(P₂)</i>	0.0352	0.0704	99.85%	0.0703	-0.19%
<i>MO(P₁)</i>	0.1204	0.1816	50.85%	0.1820	0.18%
<i>No Trade</i>	0.1835	0.1835	-0.05%	0.1831	-0.21%
<i>Quoted Spread</i>	0.0408	0.0372	-9.00%	0.0371	-0.22%
<i>BBODepth</i>	0.5106	0.7455	46.00%	0.7493	0.50%
<i>MS</i>	0.5000	0.6705	34.10%	0.6703	-0.03%

Table 3: **Equilibrium Order Submission Strategies and Market Quality in Market I. Change in MF and in TF of Market I only. 3-period vs. 4-period Model.**

This table reports for Market I the average equilibrium probabilities of the following order flows and market quality metrics: limit orders, $LO(P_i)$, and market orders, $MO(P_i)$, with the orders breakdown for the outside (P_2) and inside (P_1) price levels - for one side of the market, the other side being symmetric, Quoted Spread (*Quoted Spread*), Best Bid and Offer depth (*BBODepth*), and market share (*MS*). The table reports results obtained under two protocols (models): with 3 periods (Panel A) and 4 periods (Panel B). TF and MF for Market I are reported in rows 1 and 2 (for Market II $tf=mf=0.000$). Results are reported for different values of MF and TF, specifically: for $\{MF, TF\} = \{.00, .00\}$ (column 1), for $\{MF, TF\} = \{-.001, .001\}$ (column 2), and for $\{MF, TF\} = \{-.001, .002\}$ (column 4). Column 3 reports the percentage change ($\Delta\%$) in the market quality metrics between $\{MF, TF\} = \{.00, .00\}$ and $\{MF, TF\} = \{-.001, .001\}$ and column 5 reports the percentage change ($\Delta\%$) in the market quality metrics between $\{MF, TF\} = \{.00, .00\}$ and $\{MF, TF\} = \{-.001, .002\}$. The metrics are reported as averages across all periods of the trading game for each of the two protocols (3 and 4 periods, respectively). For both protocols, the asset value $AV = 1$ and tick size $\tau = 0.01$.

Market I TF Market I MF	Market I				
	0.000 0.000 (1)	0.001 -0.001 (2)	$\Delta\%$ (2) and (1) (3)	0.002 -0.001 (4)	$\Delta\%$ (4) and (1) (5)
Panel A. 3-period model					
<i>LO</i>	0.2521	0.0123	-95.12%	0.0123	-95.14%
<i>LO(P₂)</i>	0.0083	0.0123	48.77%	0.0123	48.49%
<i>LO(P₁)</i>	0.2438	0.0000	-99.99%	0.0000	-100.00%
<i>MO (Volume)</i>	0.1251	0.0060	-95.18%	0.0060	-95.20%
<i>MO(P₂)</i>	0.0041	0.0060	48.61%	0.0060	48.18%
<i>MO(P₁)</i>	0.1211	0.0000	-99.99%	0.0000	-100.00%
<i>No Trade</i>	0.2456	0.4063	65.41%	0.4063	65.41%
<i>Quoted Spread</i>	0.0409	0.0498	21.77%	0.0498	21.78%
<i>BBODepth</i>	0.4624	0.0214	-95.38%	0.0213	-95.39%
<i>MS</i>	0.5000	0.0241	-95.19%	0.0239	-95.21%
Panel B. 4-period model					
<i>LO</i>	0.2526	0.2502	-0.95%	0.1298	-48.60%
<i>LO(P₂)</i>	0.0682	0.2478	263.31%	0.1298	90.21%
<i>LO(P₁)</i>	0.1844	0.0024	-98.69%	0.0000	-100.00%
<i>MO (Volume)</i>	0.1556	0.0959	-38.40%	0.0521	-66.53%
<i>MO(P₂)</i>	0.0352	0.0950	169.52%	0.0521	47.82%
<i>MO(P₁)</i>	0.1204	0.0009	-99.26%	0.0000	-100.00%
<i>No Trade</i>	0.1835	0.1863	1.47%	0.1863	1.48%
<i>Quoted Spread</i>	0.0408	0.0446	9.24%	0.0472	15.69%
<i>BBODepth</i>	0.5106	0.5335	4.48%	0.2750	-46.14%
<i>MS</i>	0.5000	0.2632	-47.36%	0.1883	-62.35%

Table 4: **Trading Fee Schedules for UK and Irish listed firms.**

This table reports the trading fee schedules that apply for the LSE-listed firms during our sample period right before December 31st, 2012 to the period right after January 1st, 2013. We look at both transparent (lit) venues and dark pools. In particular, the venue that we examine are: BXE-Lit, CXE-Lit, TQ-Lit, LSE-Lit and BXE-Dark, CXE-Dark, TQ-Dark, and UBS-Dark. Our study focuses on the fee changes for the BXE-Lit and CXE-Lit markets implemented on January 1st, 2013. No other venue incurred any changes in fees.

		Effective December 31, 2012			Effective January 1, 2013		
	Tiers/Order Type	Maker fee (bps)	Taker Fee (bps)	Total Fee (bps)	Maker fee (bps)	Taker Fee (bps)	Total Fee (bps)
A. Transparent MTFs							
	BXE-Lit	-0.18	0.28	0.10	0.00	0.15	0.15
	CXE-Lit	-0.20	0.30	0.10	-0.15	0.30	0.15
	TQ-Lit						
	< €1.5bn	-0.14	0.30	0.16	-0.14	0.30	0.16
	€1.5 - €2.5bn	-0.24	0.30	0.06	-0.24	0.30	0.06
	> €2.5bn	-0.28	0.30	0.02	-0.28	0.30	0.02
B. Primary/Listing Exchange							
	LSE-Lit*						
	< £2.5bn	0.00	0.45	0.45	0.00	0.45	0.45
	£2.5 - £5.0bn	0.00	0.40	0.40	0.00	0.40	0.40
	£5.0 - £10.0bn	0.00	0.30	0.30	0.00	0.30	0.30
	> £10.0bn	0.00	0.20	0.20	0.00	0.20	0.20
C. Dark Venues							
	BXE-Dark	0.15	0.15	0.30	0.15	0.15	0.30
	CXE-Dark						
	Non-IOC Orders	0.15	0.15	0.30	0.15	0.15	0.30
	IOC Orders	0.30	0.30	0.60	0.30	0.30	0.60
	TQ-Dark	0.30	0.30	0.60	0.30	0.30	0.60
	UBS-Dark	0.10	0.10	0.20	0.10	0.10	0.20

Notes: * The 0.00 make fee only applies to passive executions qualifying under Liquidity Provider Scheme for FTSE 350 securities. LSE enforced a minimum per order charge of £0.10. Furthermore, LSE offered two Liquidity Taker Scheme Packages for Equities: 1) for a monthly fee of £50,000 the taker fee is 0.15 bps; 2) for a monthly fee of £5,000 the taker fee is 0.28 bps. Effective June 3, 2013, the hurdles for these packages were reduced to £40,000 and £4,000 respectively.

Table 5: **Descriptive Statistics for 2013 Event.**

This table reports summary statistics for our main variables. Our 120 LSE listed stocks sample is stratified by price and market capitalization, based on daily averages for the month of January 2012. All variables reported in the tables, daily measures at the stock level, are for the listing exchange only. *Volume* is defined as the daily number of shares (in 000s) at the end-of-day files from Thomson Reuters Tick History (TRTH). *Depth* is defined as the daily average of the intraday quoted depth at the ask-side and the bid-side of each quote respectively. *Spread* is defined as the time-weighted daily average of the intraday difference between the ask price and the bid price of each quote. *%Spread* is defined as the time weighted daily average of the intraday ask price minus the bid price divided by the midquote of each quote. *Volatility* is defined as the difference between the high and low trading priced of each trading day divided by the high price of that day (using the end-of-day files from TRTH). The descriptive statistics for the five measures of market quality are based on daily numbers for each stock in the one-month pre-period (December 2012). We also report *market capitalization* (in £millions) and *price* levels (in £) both variables are daily measures for the month of January 2012. In addition to the overall samples, for all of our variables we also report summary statistics for the subsamples of the highest (*Large*) and lowest (*Small*) market capitalization terciles.

Market Quality Measures		Mean	Median	ST dev	Q1	Q3
Volume (000s)	Large	10,980	3,352	23,692	1,478	7,718
	Small	767	329	1,140	119	910
	Overall	4,457	931	14,560	307	2,854
Depth	Large	11,500	7,082	16,730	4,094	11,080
	Small	6,211	1,922	14,336	867	4,882
	Overall	7,421	3,172	13,899	1,403	7,271
Spread	Large	0.898	0.722	0.812	0.215	1.486
	Small	2.050	0.891	2.748	0.369	2.658
	Overall	1.667	0.889	3.576	0.310	1.717
% Spread	Large	0.092%	0.096%	0.038%	0.060%	0.120%
	Small	0.357%	0.264%	0.330%	0.182%	0.435%
	Overall	0.228%	0.146%	0.276%	0.108%	0.246%
Volatility (High-Low)/High	Large	1.602%	1.402%	0.819%	1.101%	1.899%
	Small	2.068%	1.706%	1.387%	1.207%	2.552%
	Overall	1.886%	1.575%	1.284%	1.163%	2.211%
Market Capitalization (£Mill)	Large	20,290	8,896	24,684	4,373	25,200
	Small	789	792	169	634	926
	Overall	7,622	1,676	16,835	931	4,289
Price	Large	9.280	5.620	8.633	2.502	14.180
	Small	4.970	2.910	4.994	1.195	5.768
	Overall	6.909	4.115	6.932	2.148	9.705

Table 6: **Measures of Market Quality. Time-Series Changes for the 2013 Event.**

This table reports the changes in market quality measures (Volume (Log), Quoted Spread, Depth (Log), and Market Share) for the 2013 event using a one-month pre- and one-month post-event window. We investigate four market venues: BATS (BXE), Chi-X (CXE), Turquoise (TQ), and the primary market (LSE). Our post minus pre (differences) estimation methodology is based on running daily time-series regressions of the mean values of each measure of market quality on a dummy variable *Event* to indicate post-event period as shown in Equation (2). We run regressions for the overall sample and two subsamples of the highest (*Large*) and lowest (*Small*) market capitalization terciles. The table reports estimated coefficients and t-statistics (in parentheses) for the LSE sample. For all specifications, we employ the Newey-West correction for autocorrelation in the error terms using 10 day lags. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively

	Volume (Log)			Spread			Depth (Log)			Market Share		
	Large	Small	Overall	Large	Small	Overall	Large	Small	Overall	Large	Small	Overall
BXE												
Event	0.0914**	0.4975***	0.2558***	0.0327**	-0.5563**	-0.1829*	-0.1050***	0.0188	-0.0437***	0.0008	0.0134***	0.0075***
(t-statistic)	(2.75)	(4.08)	(3.75)	(2.54)	(-2.46)	(-1.83)	(-4.35)	(0.56)	(-3.84)	(0.70)	(3.13)	(2.94)
CXE												
Event	0.0895***	0.0424	0.0724***	0.0147***	0.1702	0.0423	-0.0309	0.0097	-0.0102	0.0040**	-0.0113***	-0.0017
(t-statistic)	(3.23)	(1.43)	(3.31)	(3.33)	(0.70)	(0.97)	(-0.77)	(0.45)	(-0.62)	(2.34)	(-5.51)	(-0.74)
TQ												
Event	0.2643***	-0.0035	0.1916**	-0.0179	-0.0505	-0.0123	0.2350***	-0.0358	0.0785***	0.0162***	-0.0054**	0.0067***
(t-statistic)	(4.42)	(-0.06)	(2.62)	(-1.17)	(-0.65)	(-0.30)	(4.62)	(-1.11)	(3.82)	(6.19)	(-2.21)	(2.62)
LSE												
Event	0.0380	0.1271***	0.0595**	0.0084	0.0666	0.0544*	0.0074	0.0576***	0.0158*	-0.0210***	0.0033	-0.0129***
(t-statistic)	(1.39)	(3.28)	(2.04)	(0.76)	(1.61)	(1.82)	(0.40)	(2.88)	(1.81)	(5.75)	(0.64)	(-3.54)

Table 7: Measures of Market Quality - Panel Regressions of the 2013 Event using ASX Sample as Control.

The table reports the changes in market quality (Volume (log), Spread, Depth (log), and Market Share) using panel difference-in-difference regressions for the 2013 event using a one-month pre- and one-month post-event window. We investigate four market venues (treatment group): BATS (BXE), Chi-X (CXE), Turquoise (TQ), and the primary market (LSE). For our control group we use a stratified sample of 120 Australian firms listed in the Australian Stock Exchange. To measure the change in market quality for each of the market venues, we follow the standard difference-in-difference specification as shown in Equation (3). The interaction variable event*treatment indicates the post-event period effect for our treatment group. We run regressions for the overall sample and two subsamples of the highest (Large) and lowest (Small) market capitalization terciles. Each Panel reports estimated coefficients and t-statistics (in parentheses) for each for the four venues. For all specifications, we employ clustered standard errors by firm and date. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively

Panel A: Panel Difference-in-difference Regressions for BXE (using ASX market as control)												
	Large	Volume (Log) Small	Overall	Large	Spread Small	Overall	Large	Depth (Log) Small	Overall	Large	Market Share Small	Overall
Intercept (t-statistic)	14.7353*** (74.50)	13.7542*** (64.89)	14.2493*** (118.12)	0.0084** (2.57)	0.0220* (1.85)	0.0124*** (7.15)	9.2607*** (33.65)	9.6809*** (31.94)	9.5454*** (56.76)	0.9047*** (119.92)	0.9192*** (141.35)	0.9122*** (208.96)
Post (t-statistic)	0.0784 (0.50)	0.0416 (0.24)	0.0084 (0.09)	0.0005 (0.14)	-0.0143 (-1.18)	-0.0075 (-1.16)	-0.0348 (-0.50)	0.0353 (0.22)	-0.0437 (-0.82)	-0.0067 (-1.02)	-0.0160** (-2.17)	-0.0128** (-2.29)
Treatment (t-statistic)	-1.8564*** (-6.73)	-4.2969*** (-11.34)	-3.1792*** (-14.19)	0.9582*** (7.00)	3.9609*** (4.53)	2.7605*** (4.19)	-1.7956*** (-5.75)	-2.9689*** (-8.66)	-2.6045*** (-13.62)	-0.8375*** (-100.73)	-0.8634*** (-98.93)	-0.8496*** (-152.22)
Post*Treatment (t-statistic)	0.0364 (0.25)	0.4568** (2.52)	0.2714*** (3.33)	0.0325*** (12.80)	-0.5222*** (-2.69)	-0.1612*** (-16.86)	-0.0745 (-1.29)	-0.0251 (-0.16)	-0.0038 (-0.08)	0.0092 (1.26)	0.0305*** (3.66)	0.0217*** (3.47)
Option Exp. Date Dummies	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
Nobs	3200	3200	9600	3192	3180	9570	3200	3200	9600	3193	3189	9581
Adj R ²	0.32	0.53	0.40	0.38	0.18	0.06	0.27	0.44	0.40	0.98	0.98	0.98
Adding the indirect and direct treatment effects:												
Post+Post*Treatment (t-statistic)	0.1147* (1.68)	0.4984*** (5.07)	0.2798*** (5.14)	0.0331 (1.04)	-0.5365*** (-2.63)	-0.1688 (-1.08)	-0.1093 (-1.43)	0.0102 (0.12)	-0.0474 (-1.02)	0.0025 (0.86)	0.0145*** (4.29)	0.0088*** (4.87)

Panel B: Panel Difference-in-difference Regressions for CXE (using ASX market as control)												
	Large	Volume (Log) Small	Overall	Large	Spread Small	Overall	Large	Depth (Log) Small	Overall	Large	Market Share Small	Overall
Intercept (t-statistic)	14.7322*** (74.59)	13.7706*** (65.01)	14.2545*** (118.24)	0.0101*** (6.54)	0.0137*** (3.28)	0.0105*** (19.84)	9.2595*** (33.64)	9.6815*** (31.94)	9.5456*** (56.76)	0.9054*** (117.03)	0.9201*** (137.77)	0.9129*** (198.94)
Post (t-statistic)	0.0805 (0.52)	0.0331 (0.19)	0.0060 (0.07)	0.0005 (0.28)	0.0030 (0.23)	-0.0040 (-0.84)	-0.0332 (-0.48)	0.0342 (0.22)	-0.0434 (-0.81)	-0.0068 (-1.01)	-0.0164** (-2.18)	-0.0131** (-2.26)
Treatment (t-statistic)	-0.5130* (-1.91)	-3.4882*** (-8.97)	-2.0943*** (-8.78)	0.7567*** (7.41)	3.6290*** (3.45)	2.3365*** (3.83)	-0.7743** (-2.53)	-2.8654*** (-8.25)	-2.0481*** (-10.26)	-0.6534*** (-64.41)	-0.8056*** (-64.97)	-0.7282*** (-80.99)
Post*Treatment (t-statistic)	0.0328 (0.23)	0.0154 (0.10)	0.0751 (1.07)	0.0142** (2.50)	0.2038*** (51.21)	0.1116*** (2.87)	-0.0032 (-0.06)	-0.0334 (-0.22)	0.0256 (0.56)	0.0132 (1.46)	0.0062 (0.75)	0.0130* (1.92)
Option Exp. Date Dummies	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
Nobs	3200	3200	9600	2192	3180	9570	3200	3200	9600	3193	3189	9581
Adj R ²	0.04	0.43	0.21	0.40	0.10	0.05	0.06	0.42	0.27	0.96	0.96	0.95
Adding the indirect and direct treatment effects:												
Post+Post*Treatment (t-statistic)	0.1133* (1.71)	0.0486 (0.48)	0.0811 (1.39)	0.0147 (0.61)	0.2067 (0.75)	0.1076 (0.70)	-0.0364 (-0.48)	0.0008 (0.01)	-0.0177 (-0.36)	0.0063* (1.95)	-0.0102** (-2.56)	-0.0001 (-0.04)

Panel C: Panel Difference-in-difference Regressions for TQ (using ASX market as control)

	Large	Volume (Log) Small	Overall	Large	Spread Small	Overall	Large	Depth (Log) Small	Overall	Large	Market Share Small	Overall
Intercept (t-statistic)	14.7357*** (74.45)	13.7683*** (65.04)	14.2553*** (118.09)	0.0086*** (2.80)	0.0016 (0.20)	0.0049 (0.86)	9.2606*** (33.65)	9.6817*** (31.94)	9.5459*** (56.77)	0.9046*** (119.98)	0.9197*** (140.46)	0.9123*** (206.88)
Post (t-statistic)	0.0782 (0.50)	0.0360 (0.21)	0.0056 (0.06)	0.0012 (0.33)	-0.0053 (-0.34)	-0.0024 (-0.24)	-0.0342 (-0.50)	0.0335 (0.21)	-0.0437 (-0.83)	-0.0066 (-1.00)	-0.0160** (-2.16)	-0.0127** (-2.26)
Treatment (t-statistic)	-2.0256*** (-7.27)	-3.9284*** (-11.09)	-3.0343*** (-14.70)	0.8976*** (7.41)	3.3707*** (4.96)	2.3792*** (4.11)	-1.8392*** (-5.89)	-3.0579*** (-8.95)	-2.6297*** (-13.80)	-0.8470*** (-98.30)	-0.8538*** (-100.99)	-0.8442*** (-145.72)
Post*Treatment (t-statistic)	0.2235 (1.50)	-0.0266 (-0.15)	0.1204 (1.30)	-0.0206*** (-23.98)	-0.0172 (-0.59)	0.0325 (0.60)	0.2659*** (4.91)	-0.0794 (-0.52)	0.1147** (2.40)	0.0243*** (3.26)	0.0111 (1.38)	0.0200*** (3.09)
Option Exp. Date Dummies	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
Nobs	3200	3200	9600	3192	3180	9568	3200	3200	9600	3193	3189	9581
Adj R ²	0.32	0.52	0.39	0.40	0.20	0.06	0.24	0.46	0.39	0.98	0.98	0.98
Adding the indirect and direct treatment effects:												
Post+Post*Treatment (t-statistic)	0.3017*** (4.22)	0.0094 (0.10)	0.1261** (2.33)	-0.0194 (-0.70)	-0.0225 (-0.13)	0.0301 (0.20)	0.2317*** (3.04)	-0.0459 (-0.54)	0.0709 (1.51)	0.0177*** (6.09)	-0.0049 (-1.47)	0.0073*** (3.98)

Panel D: Panel Difference-in-difference Regressions for LSE (using ASX market as control)

	Large	Volume (Log) Small	Overall	Large	Spread Small	Overall	Large	Depth (Log) Small	Overall	Large	Market Share Small	Overall
Intercept (t-statistic)	14.7233*** (74.71)	13.7561*** (64.97)	14.2442*** (118.40)	0.0096*** (4.76)	-0.0115 (-0.48)	-0.0038 (-0.27)	9.2560*** (33.62)	9.6773*** (31.90)	9.5425*** (56.70)	0.9020*** (120.49)	0.9173*** (135.65)	0.9100*** (199.31)
Post (t-statistic)	0.0833 (0.54)	0.0430 (0.25)	0.0114 (0.13)	0.0018 (0.89)	0.0193 (0.81)	0.0152 (1.10)	-0.0289 (-0.42)	0.0366 (0.23)	-0.0410 (-0.76)	-0.0059 (-0.86)	-0.0144* (-1.89)	-0.0117** (-2.01)
Treatment (t-statistic)	0.4036 (1.46)	-1.0956*** (-3.76)	-0.5287*** (-2.80)	0.8860*** (7.20)	2.0407*** (5.16)	1.6562*** (5.52)	-0.4282 (-1.39)	-1.9507*** (-5.37)	-1.4325*** (-7.20)	-0.2833*** (-29.60)	-0.1547*** (-8.53)	-0.2277*** (-22.35)
Post*Treatment (t-statistic)	-0.0274 (-0.20)	0.0724 (0.45)	0.0535 (0.77)	0.0051 (1.19)	0.0326* (1.76)	-0.0000 (-0.00)	0.0260 (0.46)	0.0054 (0.04)	0.0508 (1.07)	-0.0149* (-1.69)	0.0158** (2.07)	-0.0021 (-0.30)
Option Exp. Date Dummies	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
Nobs	3200	3200	9600	3190	3180	9567	3200	3200	9600	3193	3189	9581
Adj R ²	0.02	0.10	0.03	0.38	0.22	0.09	0.02	0.24	0.15	0.82	0.31	0.55
Adding the indirect and direct treatment effects:												
Post+Post*Treatment (t-statistic)	0.0559 (0.82)	0.1155 (1.45)	0.065 (1.37)	0.0069 (0.24)	0.0519 (0.53)	0.0152 (0.19)	-0.0029 (-0.04)	0.042 (0.47)	0.0098 (0.20)	-0.0208*** (-5.90)	0.0014 (0.25)	-0.0138*** (-4.51)

Table 8: Measures of Market Quality - Panel Regressions of the 2013 Event using LSE Sample as Control.

The table reports the changes in market quality (Volume (log), Spread, Depth (log), and Market Share) using panel difference-in-difference regressions for the 2013 event using a one-month pre- and one-month post-event window. We investigate four market venues (treatment group): BATS (BXE), Chi-X (CXE) and Turquoise (TQ). For our control group we use the LSE. To measure the change in market quality for each of the market venues, we follow the standard difference-in-difference specification as shown in Equation (4). The interaction variable event*treatment indicates the post-event period effect for our treatment group. Since we expect indirect effects on the LSE market due to intermarket competition, and following [Boehmer et al. \(2020\)](#) we also report the joint direct and indirect effects (event+event*treatment). We run regressions for the overall sample and two subsamples of the highest (Large) and lowest (Small) market capitalization terciles. Each Panel reports estimated coefficients and t-statistics (in parentheses) for each for the three venues (BXE, CXE, and TQ). For all specifications, we employ clustered standard errors by firm times date. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

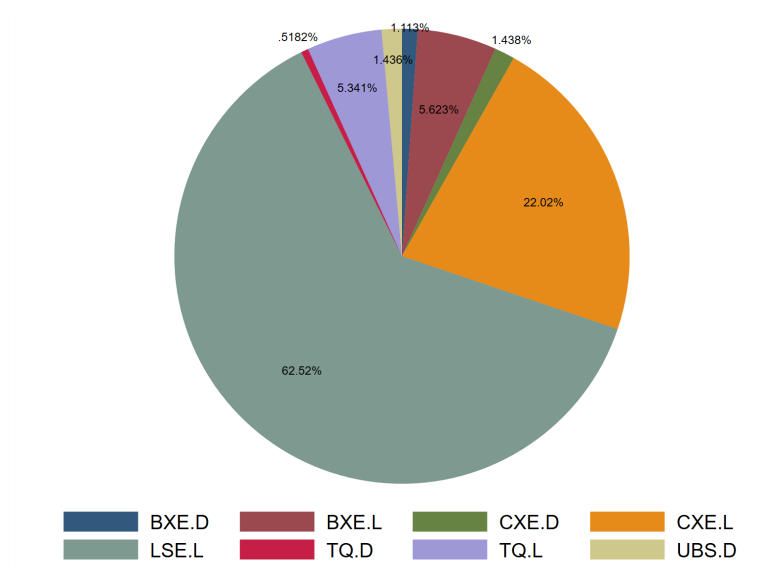
Panel A: 2013 Event for LSE sample – Panel Difference-in-difference Regressions for BXE (using LSE market as control)												
	Volume (Log)			Spread			Depth (Log)			Market Share		
	Large	Small	Overall	Large	Small	Overall	Large	Small	Overall	Large	Small	Overall
Intercept (t-statistic)	15.1447*** (317.50)	12.6536*** (243.73)	13.7196*** (376.07)	0.8924*** (30.23)	2.0417*** (20.42)	1.6542*** (21.43)	8.8244*** (230.59)	7.7179*** (163.17)	8.1044*** (309.20)	0.6202*** (272.41)	0.7630*** (157.56)	0.6832*** (271.04)
Post (t-statistic)	0.0327 (0.49)	0.1256* (1.67)	0.0604 (1.18)	0.0074 (0.18)	0.0333 (0.24)	0.0025 (0.02)	0.0016 (0.03)	0.0559 (0.86)	0.0164 (0.45)	-0.0219*** (-7.34)	0.0016 (0.25)	-0.0136*** (-3.83)
Treatment (t-statistic)	-2.2600*** (-159.12)	-3.2014*** (-47.25)	-2.6506*** (-99.28)	0.0722*** (6.36)	1.9201*** (12.64)	1.0744*** (11.45)	-1.3674*** (-57.55)	-1.0182*** (-50.07)	-1.1719*** (-92.51)	-0.5541*** (-194.17)	-0.7086*** (-117.98)	-0.6219*** (-206.37)
Post*Treatment (t-statistic)	0.0639*** (3.46)	0.3743*** (4.29)	0.2105*** (6.18)	0.0264* (1.78)	-0.5563*** (-2.98)	-0.1530 (-1.04)	-0.1009 (-3.37)	-0.0356 (-1.27)	-0.0560*** (-2.82)	0.0236*** (6.63)	0.0134 (1.59)	0.0225*** (5.33)
Option Exp. Date Dummies	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
Adding the indirect and direct treatment effects:												
Post+Post*Treatment (t-statistic)	0.0966 (1.44)	0.4990*** (4.49)	0.2709*** (4.25)	0.0338 (0.76)	-0.5230* (-1.85)	-0.1505 (-0.69)	-0.0993** (-1.99)	0.0203 (0.37)	-0.0396 (-1.30)	0.0017 (1.54)	0.0151*** (6.47)	0.0089*** (8.35)

Panel B: 2013 Event for LSE sample – Panel Difference-in-difference Regressions for CXE (using LSE market as control)												
	Volume (Log)			Spread			Depth (Log)			Market Share		
	Large	Small	Overall	Large	Small	Overall	Large	Small	Overall	Large	Small	Overall
Intercept (t-statistic)	15.1416*** (318.01)	12.6700*** (241.83)	13.7249*** (374.38)	0.8941*** (30.50)	2.0334*** (20.20)	1.6523*** (21.61)	8.8231*** (230.88)	7.7184*** (163.03)	8.1046*** (308.13)	0.6210*** (270.40)	0.7639*** (158.15)	0.6839*** (271.67)
Post (t-statistic)	0.0349 (0.52)	0.1168 (1.54)	0.0579 (1.13)	0.0073 (0.18)	0.0493 (0.35)	0.0059 (0.11)	0.0031 (0.06)	0.0550 (0.85)	0.0167 (0.46)	-0.0221*** (-7.37)	0.0013 (0.19)	-0.0139*** (-3.92)
Treatment (t-statistic)	-0.9166*** (-88.53)	-2.3926*** (-44.23)	-1.5656*** (-65.01)	-0.1293*** (-14.51)	1.5882*** (7.87)	0.6804*** (7.27)	-0.3461*** (-17.08)	-0.9147*** (-43.66)	-0.6156*** (-49.87)	-0.3700*** (-96.24)	-0.6508*** (-88.61)	-0.5005*** (-119.22)
Post*Treatment (t-statistic)	0.0578*** (4.22)	-0.0640 (-0.85)	0.0153 (0.45)	0.0086 (0.71)	0.1384 (0.46)	0.1161 (0.75)	-0.0299 (-1.05)	-0.0427 (-1.41)	-0.0269 (-1.57)	0.0272*** (5.37)	-0.0106 (-1.05)	0.0136** (2.30)
Option Exp. Date Dummies	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
Adding the indirect and direct treatment effects:												
Post+Post*Treatment (t-statistic)	0.0927 (1.46)	0.0527 (0.46)	0.0732 (1.05)	0.0159 (0.47)	0.1876 (0.49)	0.1219 (0.57)	-0.0267 (-0.57)	0.0123 (0.22)	-0.0102 (-0.28)	0.0051** (2.22)	-0.0093** (-2.52)	-0.0003 (-0.10)

Panel C: 2013 Event for LSE sample – Panel Difference-in-difference Regressions for TQ (using LSE market as control)												
	Volume (Log)			Spread			Depth (Log)			Market Share		
	Large	Small	Overall	Large	Small	Overall	Large	Small	Overall	Large	Small	Overall
Intercept (t-statistic)	15.1451*** (317.31)	12.6677*** (243.60)	13.7257*** (376.99)	0.8926*** (30.32)	2.0215*** (20.44)	1.6468*** (21.48)	8.8243*** (230.55)	7.7187*** (163.22)	8.1049*** (309.08)	0.6202*** (272.60)	0.7634*** (157.63)	0.6834*** (270.86)
Post (t-statistic)	0.0322 (0.48)	0.1194 (1.58)	0.0571 (1.12)	0.0080 (0.20)	0.0423 (0.30)	0.0077 (0.07)	0.0021 (0.04)	0.0543 (0.84)	0.0164 (0.45)	-0.0218*** (-7.31)	0.0016 (0.25)	-0.0135*** (-3.81)
Treatment (t-statistic)	-2.4293*** (-154.99)	-2.8329*** (-55.62)	-2.5056*** (-119.48)	0.0116 (1.56)	1.3300*** (13.38)	0.7231*** (9.28)	-1.4109*** (-60.95)	-1.1073*** (-56.61)	-1.1972*** (-102.85)	-0.5636*** (-197.27)	-0.6990*** (-113.87)	-0.6165*** (-203.06)
Post*Treatment (t-statistic)	0.2379*** (6.30)	-0.1152 (-1.55)	0.0439 (1.24)	-0.0254** (-2.44)	-0.0686 (-0.49)	0.0410 (0.30)	0.2382*** (7.71)	-0.0869*** (-3.04)	0.0633*** (3.91)	0.0389*** (10.28)	-0.0053 (-0.63)	0.0211*** (4.90)
Option Exp. Date Dummies	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
Adding the indirect and direct treatment effects:												
Post+Post*Treatment (t-statistic)	0.2701*** (3.60)	0.0042 (0.04)	0.1010 (1.60)	-0.0174 (-0.45)	-0.0262 (0.23)	0.0487 (0.24)	0.2403*** (4.82)	-0.0326 (-0.61)	0.0797** (2.55)	0.0170*** (13.09)	-0.0037* (-1.68)	0.0075*** (6.55)

Figure 1: **Market Share Pie-Charts of the LSE sample in 2012 (Pre-Event) and 2015 (Post-Event)**
The pie-chart figures show average daily market share of each market venue used in the analysis for the LSE sample in the pre-period of the 2013 event (November and December 2012) and in the period after fee change in January 2013 (February and March 2015). In particular, we look at both lit markets (LSE.L, CXE.L, BXE.L, and TQ.L) and dark pool venues (CXE.D, BXE.D, TQ.D, and UBS.D) market share. We exclude other trading venues and off-market trades for the pie-charts. Market share data were collected from Fidessa (Fragulator).

(a) 2013 Event Pre-Period (November and December 2012)



(b) 2013 Event Post-Period (February and March 2013)

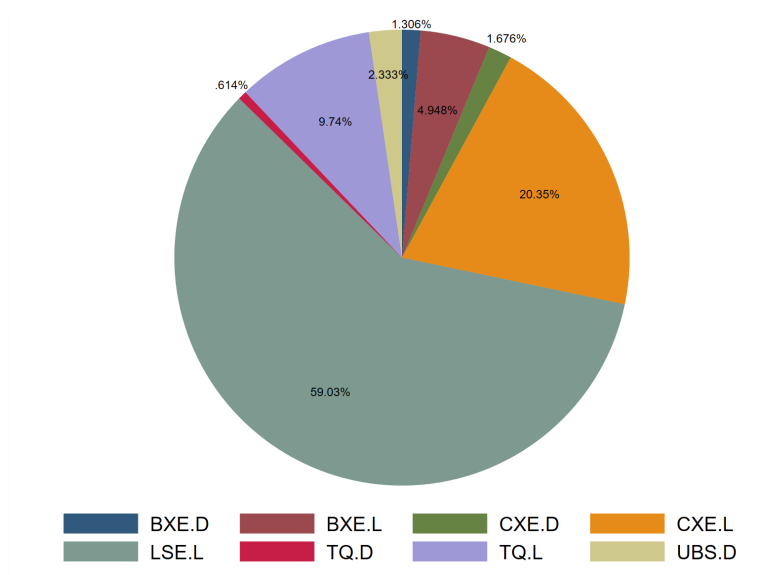
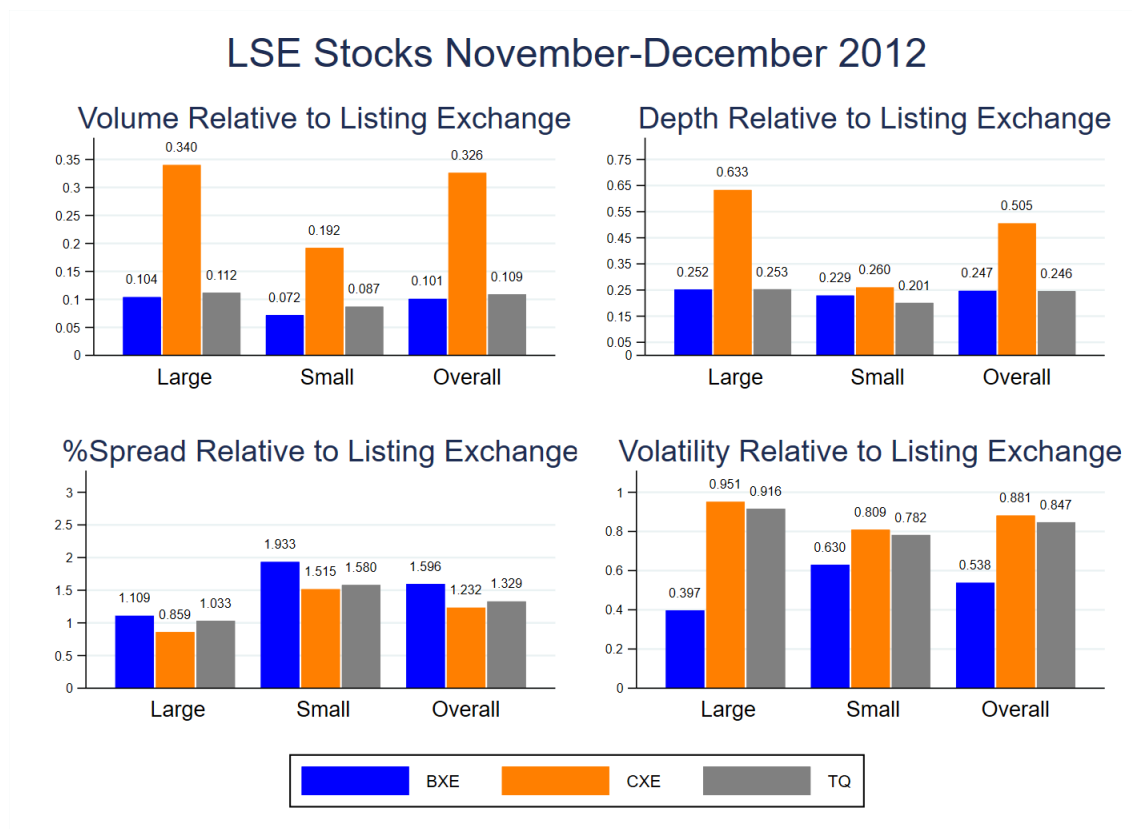


Figure 2: Market Quality Measures across Markets

The figure shows average daily market quality measures (Volume, Depth, %Spread, and Volatility) of the three market venues (BXE, CXE, TQ) relative to the listing exchange (LSE) in the pre-period (Nov/Dec 2012) of the 2013 Event. It depicts relative market quality measures for the overall sample and two sub-samples of the highest (Large) and lowest (Small) market capitalization terciles. Filled bars indicate that a venue mean is significantly different from the listing exchange mean based on a simple differences-in-means test.



Appendix 1

Model Solution and proof of Proposition 1 and 2

In this Appendix we show how to solve the 3-period model; the solution of the 4-period model follows the same line of reasoning. At each period t_z , a trader uses the information from the state of the book of both Market I and Market II to rationally compute and compare the payoffs from the available strategies (Table 1). However, to compare the payoffs across these strategies, the trader has to compute the execution probabilities of limit orders, which are uncertain as they depend on the probability of the t_{z+1} (and possibly t_{z+2}) market order submissions. To overcome this issue, the model is solved by backward induction starting from the last period of the trading game, t_3 . At t_3 the execution probabilities of limit orders, $LO_{t_3}(P_i^j)$, are equal to zero and therefore to choose the order submission strategy ($ST_{t_3}^*$) that maximizes the expected payoff ($\pi_{t_3}^e$) conditional on their private valuation of the asset, γ , traders solve problem (5) by choosing between market orders, $MO_{t_3}(P_i^{j,b})$, and no-trade $NT_{t_3}(0)$:

$$\max_{ST_{t_3}^*} \pi_{t_3}^e \left\{ MO_{t_3}(P_i^{j,b}), NT_{t_3}(0) \mid \gamma, lob_{t_3}^j \right\} \quad (5)$$

Table 1 shows that the non-zero traders' payoffs are a function of $\gamma \in (\bar{\gamma}, \underline{\gamma})$. We can therefore rank the payoffs of adjacent optimal strategies in terms of γ and equate them to determine the t_3 equilibrium γ thresholds in the following way:

$$\gamma_{t_3}^{ST_n^*, ST_{n-1}^*} = \left\{ \gamma \in \mathbb{R} : \pi_{t_3}^e \left(ST_n^* \mid lob_{t_3}^j \right) - \pi_{t_3}^e \left(ST_{n-1}^* \mid lob_{t_3}^j \right) = 0 \right\} \quad (6)$$

By using the γ thresholds together with the cumulative distribution function (CDF) of γ , $F(\cdot)$, we can now derive the probability of each equilibrium order submission strategy, $ST_{\cdot}^*$, conditional on all the possible combinations of the t_3 states of the book:

$$Pr[ST_n^* \mid lob_{t_3}^j] = F(\gamma_{t_3}^{ST_{n+1}^*, ST_n^*} \mid lob_{t_3}^j) - F(\gamma_{t_3}^{ST_n^*, ST_{n-1}^*} \mid lob_{t_3}^j) \quad (7)$$

Clearly, the probability to observe a $MO_{t_3}(P_i^{j,b})$ at t_3 is the execution probability of a $LO_{t_2}(P_i^j)$ at t_2 , therefore, we can now compute and compare the t_2 payoffs to determine the equilibrium γ thresholds and therefore the equilibrium order submission probabilities conditional on each possible combination of the states of the book in the two markets at t_2 . The t_1 equilibrium order submission strategies can then be recursively obtained, as the t_2 market orders' equilibrium probabilities are the execution probabilities of the limit orders posted at t_1 .

As a general example, consider a case at t_3 with the book that opens empty and with one sell order standing on the first level of Market II and one buy order standing on the second level of Market I. This means that the payoffs from the t_3 strategies are:

$$\begin{aligned}
\pi_{t_3}^e(MO_{t_3}(S_1^{mrkII}) | lob_{t_3}^j) &= \gamma AV - S_1^{mrkII} - tf \\
\pi_{t_3}^e(NT_{t_3}(0) | lob_{t_3}^j) &= 0 \\
\pi_{t_3}^e(MO_{t_3}(B_2^P) | lob_{t_3}^j) &= B_2^{mrkI} - \gamma AV - TF
\end{aligned} \tag{8}$$

Hence the t_3 equilibrium strategies are:

$$ST_{(\cdot)}^* = \begin{cases} MO_{t_3}(S_1^{mrkII}) & \text{if } \gamma \in [\underline{\gamma}, \frac{S_1^{mrkII} - tf}{AV}) \\ NT_{t_3}(0) & \text{if } \gamma \in [\frac{S_1^{mrkII} - tf}{AV}, \frac{B_2^{mrkI} + TF}{AV}) \\ MO_{t_3}(B_2^{mrkI}) & \text{if } \gamma \in (\frac{B_2^{mrkI} + TF}{AV}, \bar{\gamma}] \end{cases} \tag{9}$$

and the t_3 equilibrium order submission probabilities are:

$$Pr[ST_{(\cdot)}^* | lob_{t_3}^j] = \begin{cases} \int_{\gamma \in \{\gamma: ST_{(\cdot)}^* = MO_{t_3}(1, S_1^{mrkII})\}} g(\gamma) d\gamma \\ \int_{\gamma \in \{\gamma: ST_{(\cdot)}^* = NT_{t_3}(0)\}} g(\gamma) d\gamma \\ \int_{\gamma \in \{\gamma: ST_{(\cdot)}^* = MO_{t_3}(1, B_2^{mrkI})\}} g(\gamma) d\gamma \end{cases} \tag{10}$$

where $g(\gamma)$ is the probability density function (PDF) of γ .

Note that $Pr[MO_{t_3}(S_1^{mrkII}) | lob_{t_3}^j]$ and $Pr[MO_{t_3}(B_2^{mrkI}) | lob_{t_3}^j]$ correspond to the execution probabilities of the previous period (t_2) limit orders respectively posted to Market II and Market I, i.e., $[LO_{t_2}(S_1^{mrkII}) | lob_{t_2}^j]$ and $[LO_{t_2}(B_2^{mrkI}) | lob_{t_2}^j]$, which are the dynamic link between periods t_3 and t_2 .

As an example, we now solve the model to obtain the results shown in Table A7 for one set of trading fees: $\{MF, TF\} = \{-.001, .001\}$ and $\{mf, tf\} = \{.00, .00\}$. Results for the other sets of fees can be obtained in a similar way. Tables A1, A2 and A3 show the equilibrium strategies (column 1) at t_3 , t_2 and t_1 respectively for all the possible states of the book starting from an empty book at t_1 . Each table also shows the payoff associated to each equilibrium strategy (column 2), the γ thresholds indicating the corresponding support of the TN distribution for each equilibrium strategy (column 3), and the resulting submission probabilities (column 4).⁴⁵

The model is solved by backward induction, so as an example, following the branch of the trading game that starts at t_1 with $LO_{t_1}(S_1^{mrkII})$, the book opens at t_2 as [0000-0100].⁴⁶ Given the three equilibrium strategies that result when we condition to this opening book at t_2 , $[NT_{t_2}(0), LO_{t_2}(B_2^{mrkI}) \text{ and } MO(S_1^{mrkII})]$, at t_3 the book may open with three different states, [0000-0100], [0001-0100], and [0000-0000], respectively. The last column of each table shows the submission probability of the equilibrium orders which are then used to compute

⁴⁵The γ thresholds indicate the optimal trading strategies that result from comparing the payoffs of all the possible orders a trader can choose conditional on each state of the book in any trading period (Equation (6)).

⁴⁶[0000-0100] indicates the state of Market I and Market II respectively, $[l^{S_2^{mrkI}} l^{S_1^{mrkI}} l^{B_1^{mrkI}} l^{B_2^{mrkI}} l^{S_2^{mrkII}} l^{S_1^{mrkII}} l^{B_1^{mrkII}} l^{B_2^{mrkII}}]$.

both the metrics of order flows (average limit orders, LO^j , and average market share MS^j), and the metrics of market quality, (average quoted spread, $Spread^j$ and average depth at the best bid-offer, $BBODepth^j$), shown in Table 3 for the above mentioned set of trading fees: $\{MF, TF\} = \{-.001, .001\}$ and $\{mf, tf\} = \{.00, .00\}$. Finally, Tables A4 and A5 show how to obtain both the order flows and the market quality metrics for this set of fees, starting from the equilibrium order submission strategies. Therefore, Tables A4 and A5 link Tables A1, A2 and A3 with Table 3.⁴⁷ Results for different sets of fees can be obtained in a similar way.

Effects of change in the number of trading periods

Table 3 (and Table A7) reports results for both the 3-period and the 4-period models. This way we can investigate how our results change when the market is characterized by a different trading activity. To understand this consider the results for the equilibrium order submission probability of both limit and market order submissions, as well as the derived metrics of market quality reported in column 3 of Tables 3 and A7. These results are obtained by solving the model for the regime with all the trading fees set equal to zero, $\{MF, TF, mf, tf\} = \{.00, .00, .00, .00\}$.⁴⁸

All else equal, when the trading activity increases from $N = 3$ to $N = 4$, the execution probability of limit orders increases as orders have an additional period to execute. As a consequence, some limit orders move from the inside to the outside quotes and overall liquidity supply increases.⁴⁹ This explains why $BBODepth^{mrkI}$ and $Quoted Spread^{mrkI}$ improve.

⁴⁷Results for average values reported in Tables A4 and A5 have been obtained by rounding at the fourth decimal value and they may slightly differ from the results reported in column 2 of Table 3 which have been obtained without any rounding.

⁴⁸Appendix 1 shows how the metrics of market quality are obtained starting from the equilibrium order submission probabilities.

⁴⁹Note that even if the average order submission probability of limit orders across the trading game decreases in the 4-period model compared to the 3-period one, liquidity provision overall increases in the 4-period protocol. The reason is that as the book fills up with limit orders, over time there is less room for traders to post additional limit orders; therefore, even though in the first two periods of the trading game the average order submission probability of limit orders in the 4-period model increases compared to the 3-period model, as the book fills up with limit orders, in the additional third period, t_3 , the average probability of limit order submission decreases, with the consequence that the overall average of limit order submission probability in the 4-period model decreases.

Table A1: **Equilibrium Strategies at t_3** This table shows how to derive the equilibrium order submission strategies at t_3 of the 3-period model - for the following set of trading fees: $\{MF, TF\} = \{-.001, .001\}$ and $\{mf, tf\} = \{.00, .00\}$ and for $\gamma \in ([0.0, 2.0])$. At t_1 both Market I and Market II open with an empty book, $[0000-0000]$, where each element in the square bracket, $l_{t_z}^{S_z^j}$, corresponds to the depth of the book at each price level of both Market I and Market II at time t_z , $[l_{t_z}^{S_1^{mrkI}} l_{t_z}^{S_1^{mrkII}} l_{t_z}^{B_1^{mrkI}} l_{t_z}^{B_1^{mrkII}} - l_{t_z}^{S_2^{mrkI}} l_{t_z}^{S_2^{mrkII}} l_{t_z}^{B_2^{mrkI}} l_{t_z}^{B_2^{mrkII}}]_{t_z}$. Given the chosen set of fees, four are the equilibrium strategies at t_1 , $LO_{t_1}(S_1^{mrkII})$, $LO_{t_1}(S_2^{mrkI})$, $LO_{t_1}(B_1^{mrkII})$ and $LO_{t_1}(B_2^{mrkI})$. We only consider the sell side of the market, the buy side being symmetrical. Given the equilibrium limit sell orders, the possible states of the books at the beginning of t_2 are: $[0000-0100]$ and $[1000-0000]$. Given the equilibrium strategies at t_2 and therefore the possible states of the books at the beginning of t_3 , this table shows the equilibrium Strategies at t_3 (column 1), their payoffs (column 2), the γ thresholds (column 3) and the order submission probabilities (column 4).

Equilibrium Strategy	Payoff	γ Threshold	Order Submission Probability
at t_1 mrkI and mrkII books open empty $[0000-0000]$: equilibrium strategy $LO_{t_1}(S_1^{mrkII})$			
at t_2 mrkI and mrkII books open $[0000-0100]$			
t_2 equilibrium strategy NT_{t_2}			
at t_3 mrkI and mrkII books open $[0000-0100]$			
$NT_{t_3}(0)$	0	$\{0.0000, 1.0050\}$	0.5025
$MO_{t_3}(S_1^{mrkII})$	$\gamma AV - S_1^{mrkII} - tf = \gamma - 1.0050$	$\{1.0050, 2.0000\}$	0.4975
t_2 equilibrium strategy $LO_{t_2}(B_2^{mrkI})$			
at t_3 mrkI and mrkII books open $[0001-0100]$			
$MO_{t_3}(B_2^{mrkI})$	$B_2^{mrkI} - \gamma AV - TF = 0.9840 - \gamma$	$\{0.0000, 0.9840\}$	0.4920
$NT_{t_3}(0)$	0	$\{0.9840, 1.0050\}$	0.0105
$MO_{t_3}(S_1^{mrkII})$	$\gamma AV - S_1^{mrkII} - tf = \gamma - 1.0050$	$\{1.0050, 2.0000\}$	0.4975
t_2 equilibrium strategy $MO_{t_2}(S_1^{mrkII})$			
at t_3 mrkI and mrkII books open $[0000-0000]$			
$NT_{t_3}(0)$	0	$\{0.0000, 2.0000\}$	1.0000
at t_1 mrkI and mrkII books open empty $[0000-0000]$: equilibrium strategy $LO_{t_1}(S_2^{mrkI})$			
at t_2 mrkI and mrkII books open $[1000-0000]$			
t_2 equilibrium strategy $LO_{t_2}(S_1^{mrkII})$			
at t_3 mrkI and mrkII books open $[1000-0100]$			
$NT_{t_3}(0)$	0	$\{0.0000, 1.0050\}$	0.5025
$MO_{t_3}(S_1^{mrkII})$	$\gamma AV - S_1^{mrkII} - tf = \gamma - 1.0050$	$\{1.0050, 2.0000\}$	0.4975
at t_2 equilibrium strategy $LO_{t_2}(S_1^{mrkI})$			
at t_3 mrkI and mrkII books open $[1100-0000]$			
$NT_{t_3}(0)$	0	$\{0.0000, 1.0060\}$	0.5030
$MO_{t_3}(S_1^{mrkI})$	$\gamma AV - S_1^{mrkI} - TF = \gamma - 1.0060$	$\{1.0060, 2.0000\}$	0.4970
at t_2 equilibrium strategy $LO_{t_2}(S_2^{mrkII})$			
at t_3 mrkI and mrkII books open $[1000-1000]$			
$NT_{t_3}(0)$	0	$\{0.0000, 1.0150\}$	0.5075
$MO_{t_3}(S_2^{mrkII})$	$\gamma AV - S_2^{mrkII} - tf = \gamma - 1.0150$	$\{1.0150, 2.0000\}$	0.4925
at t_2 equilibrium strategy $LO_{t_2}(B_2^{mrkI})$			
at t_3 mrkI and mrkII books open $[1001-0000]$			
$MO_{t_3}(B_2^{mrkI})$	$B_2^{mrkI} - \gamma AV - TF = 0.9840 - \gamma$	$\{0.0000, 0.9840\}$	0.4920
$NT_{t_3}(0)$	0	$\{0.9840, 1.0160\}$	0.0160
$MO_{t_3}(S_2^{mrkI})$	$\gamma AV - S_2^{mrkI} - TF = \gamma - 1.0160$	$\{1.0160, 2.0000\}$	0.4920
at t_2 equilibrium strategy $MO_{t_2}(S_2^{mrkI})$			
at t_3 mrkI and mrkII books open $[0000-0000]$			
$NT_{t_3}(0)$	0	$\{0.0000, 2.0000\}$	1.0000

Table A2: **Equilibrium Strategies at t_2** This table shows how to derive the equilibrium order submission strategies at t_2 of the 3-period model - for the following set of trading fees: $\{MF, TF\} = \{-.001, .001\}$ and $\{mf, tf\} = \{.00, .00\}$ and for $\gamma \in [0.0, 2.0]$. At t_1 both Market I and Market II open with an empty book, [0000-0000], where each element in the square bracket, $l_{t_z}^{S_i^j}$, corresponds to the depth of the book at each price level of both Market I and Market II at time t_z , $[l_{t_z}^{S_2^{mrkI}} l_{t_z}^{S_1^{mrkI}} l_{t_z}^{B_1^{mrkI}} l_{t_z}^{B_2^{mrkI}} - l_{t_z}^{S_2^{mrkII}} l_{t_z}^{S_1^{mrkII}} l_{t_z}^{B_1^{mrkII}} l_{t_z}^{B_2^{mrkII}}]_{t_z}$. Given the chosen set of fees, four are the equilibrium strategies at t_1 , $LO_{t_1}(S_1^{mrkII})$, $LO_{t_1}(S_2^{mrkI})$, $LO_{t_1}(B_1^{mrkII})$ and $LO_{t_1}(B_2^{mrkI})$. We only consider the sell side of the market, the buy side being symmetrical. Given the equilibrium limit sell orders, the possible states of the books at the beginning of t_2 are: [0000-0100] and [1000-0000]. Column 1 shows the Equilibrium strategies at t_2 , column 2 shows their payoffs, and columns 3 and 4 shows the γ thresholds and the order submission probabilities respectively.

Equilibrium Strategy	Payoff	γ Threshold	Order Submission Probability
at t_1 mrkI and mrkII books open empty [0000-0000]: equilibrium strategy $LO_{t_1}(S_1^{mrkII})$			
at t_2 mrkI and mrkII books open [0000-0100]			
$NT_{t_2}(0)$	0	{0.0000, 0.9840}	0.4920
$LO_{t_2}(B_2^{mrkI})$	$(\gamma AV - B_2^{mrkI} - MF) \times Pr(MO_{t_3}(B_2^{mrkI})) [0001 - 0100] = -0.4841 + 0.4920\gamma$	{0.9840, 1.0253}	0.0207
$MO_{t_2}(S_1^{mrkII})$	$\gamma AV - S_1^{mrkII} - tf = -1.005 + \gamma$	{1.0253, 2.0000}	0.4873
at t_1 mrkI and mrkII books open empty [0000-0000]: equilibrium strategy $LO_{t_1}(S_2^{mrkI})$			
at t_2 mrkI and mrkII books open [1000-0000]			
$LO_{t_2}(S_1^{mrkII})$	$(S_1^{mrkII} - \gamma AV - mf) \times Pr(MO_{t_3}(S_1^{mrkII})) [1000 - 0100] = 0.49998 - 0.4975\gamma$	{0.0000, 0.0110}	0.0055
$LO_{t_2}(S_1^{mrkI})$	$(S_1^{mrkI} - \gamma AV - MF) \times Pr(MO_{t_3}(S_1^{mrkI})) [1100 - 0000] = 0.49998 - 0.4970\gamma$	{0.0110, 0.0210}	0.0050
$LO_{t_2}(S_2^{mrkII})$	$(S_2^{mrkII} - \gamma AV - mf) \times Pr(MO_{t_3}(S_2^{mrkII})) [1000 - 1000] = 0.49989 - 0.4925\gamma$	{0.0210, 0.9995}	0.4893
$LO_{t_2}(B_2^{mrkI})$	$(\gamma AV - B_2^{mrkI} - MF) \times Pr(MO_{t_3}(B_2^{mrkI})) [1001 - 0000] = -0.48413 + 0.4920\gamma$	{0.9995, 1.0470}	0.0237
$MO_{t_2}(S_2^{mrkI})$	$\gamma AV - S_2^{mrkI} - TF = -1.016 + \gamma$	{1.0470, 2.0000}	0.4765

Table A3: **Equilibrium Strategies at t_1** This table shows how to derive the equilibrium order submission strategies at t_1 - of the 3-period model - for the following set of trading fees: $\{MF, TF\} = \{-.001, .001\}$ and $\{mf, tf\} = \{.00, .00\}$ and for $\gamma \in [0.0, 2.0]$. At t_1 both Market I and Market II open with an empty book, [0000-0000], where each element in the square bracket, $l_{t_z}^{S_i^j}$, corresponds to the depth of the book at each price level of both Market I and Market II at time t_z , $[l_{t_z}^{S_2^{mrkI}} l_{t_z}^{S_1^{mrkI}} l_{t_z}^{B_1^{mrkI}} l_{t_z}^{B_2^{mrkI}} - l_{t_z}^{S_2^{mrkII}} l_{t_z}^{S_1^{mrkII}} l_{t_z}^{B_1^{mrkII}} l_{t_z}^{B_2^{mrkII}}]_{t_z}$. Given the chosen set of fees, four are the equilibrium strategies at t_1 , $LO_{t_1}(S_1^{mrkII})$, $LO_{t_1}(S_2^{mrkI})$, $LO_{t_1}(B_1^{mrkII})$ and $LO_{t_1}(B_2^{mrkI})$ which are shown in column 1. Column 2 shows their payoffs, and column 3 and 4 shows the γ thresholds and the order submission probabilities respectively.

Equilibrium Strategy	Payoff	γ Threshold	Order Submission Probability
at t_1 mrkI and mrkII books open empty [0000-0000]			
$LO_{t_1}(S_1^{mrkII})$	$(S_1^{mrkII} - \gamma AV - mf) \times [(Pr(MO_{t_2}(S_1^{mrkII})) [0000 - 0100]) + (1 - Pr(MO_{t_2}(S_1^{mrkII})) [0000 - 0100]) \times Pr(MO_{t_3}(S_1^{mrkII})) [0000 - 0100])] = 0.7461 - 0.7424\gamma$	{0.0000, 0.9839}	0.4919
$LO_{t_1}(S_2^{mrkI})$	$(S_2^{mrkI} - \gamma AV - MF) \times [(Pr(MO_{t_2}(S_2^{mrkI})) [1000 - 0000]) + (1 - Pr(MO_{t_2}(S_2^{mrkI})) [1000 - 0000]) \times Pr(MO_{t_3}(S_2^{mrkI})) [1000 - 0000])] = 0.4960 - 0.4882\gamma$	{0.9839, 1.0000}	0.0081
$LO_{t_1}(B_2^{mrkI})$	$(\gamma AV - B_2^{mrkI} - MF) \times [(Pr(MO_{t_2}(B_2^{mrkI})) [0001 - 0000]) + (1 - Pr(MO_{t_2}(B_2^{mrkI})) [0001 - 0000]) \times Pr(MO_{t_3}(B_2^{mrkI})) [0001 - 0000])] = -0.4804 + 0.4882\gamma$	{1.0000, 1.0161}	0.0081
$LO_{t_1}(B_1^{mrkII})$	$(\gamma AV - B_1^{mrkII} - mf) \times [(Pr(MO_{t_2}(B_1^{mrkII})) [0000 - 0010]) + (1 - Pr(MO_{t_2}(B_1^{mrkII})) [0000 - 0010]) \times Pr(MO_{t_3}(B_1^{mrkII})) [0000 - 0010])] = -0.7387 + 0.7424\gamma$	{1.0161, 2.0000}	0.4919

Table A4: **Equilibrium Order Submission Strategies, Order Flows and Market Quality** This table shows how to obtain the metrics on order flows and market quality (column 2) presented in Tables 3 and A7 for the following set of trading fees: $\{MF, TF\} = \{-.001, .001\}$ and $\{mf, tf\} = \{.00, .00\}$. Column 2 reports the equilibrium order submission probability of limit and market orders Market I and Market II in each period t_z , $LO_{t_z}^j$ and $MO_{t_z}^j$, and the equilibrium average of limit orders, market orders and market share, LO^j , MO^j and MS^j . Column 3 shows how the values reported in column 2 are computed from the equilibrium strategies. Results are reported for both Market I (mrkI) and Market II (mrkII) market. Traders have a personal evaluation of the asset which is a Uniform distribution, $\gamma \sim U[\bar{\gamma}, \underline{\gamma}]$ with $S = [0, 2]$ and $AV = 1$.

Metric	Value	Analytical Computation
$LO_{t_1}^{mrkI}$	0.0162	$Pr(LO_{t_1}(S_2^{mrkI}) .) + Pr(LO_{t_1}(B_2^{mrkI}) .) = 0.0081 + 0.0081$
$LO_{t_2}^{mrkI}$	0.0208	$(2 \times Pr(LO_{t_1}(S_1^{mrkII}) .) \times Pr(LO_{t_2}(B_2^{mrkI}) 0000 - 0100) + (2 \times Pr(LO_{t_1}(S_2^{mrkI}) .) \times (Pr(LO_{t_2}(S_1^{mrkI}) 1000 - 0000) + Pr(LO_{t_2}(B_2^{mrkI}) 1000 - 0000))$ $= (2 \times 0.4919) \times 0.0207 + (2 \times 0.0081) \times (0.0050 + 0.0237)$
LO^{mrkI}	0.0185	$(LO^{mrkI}_{t_1} + LO^{mrkI}_{t_2})/2 = (0.0162 + 0.0208)/2$
$MO_{t_2}^{mrkI}$	0.0077	$2 \times Pr(LO_{t_1}(S_2^{mrkI}) .) \times Pr(MO_{t_2}(S_2^{mrkI}) 1000 - 0000) = 2 \times 0.0081 \times 0.4765$
$MO_{t_3}^{mrkI}$	0.0104	$(2 \times Pr(LO_{t_1}(S_2^{mrkI}) .) \times (Pr(LO_{t_2}(S_1^{mrkI}) 1000 - 0000) \times Pr(MO_{t_3}(S_1^{mrkI}) 1100 - 0000) + Pr(LO_{t_2}(B_2^{mrkI}) 1001 - 0000)$ $\times (Pr(MO_{t_3}(S_2^{mrkI}) 1001 - 0000) + Pr(MO_{t_3}(B_2^{mrkI}) 1001 - 0000))) + (2 \times Pr(LO_{t_1}(S_1^{mrkII}) .) \times Pr(LO_{t_2}(B_2^{mrkI}) 0000 - 0100) \times Pr(MO_{t_3}(B_2^{mrkI}) 0001 - 0100))$ $= 2 \times 0.0081 \times (0.0050 \times 0.4970 + 0.0237 \times (0.492 + 0.492)) + 2 \times 0.4919 \times 0.0207 \times 0.4920$
MO^{mrkI}	0.0091	$(MO_{t_2}^{mrkI} + MO_{t_3}^{mrkI})/2 = (0.0077 + 0.0104)/2$
MS^{mrkI}	0.0242	$MO^{mrkI} / (MO^{mrkI} + MO^{mrkII}) = 0.0091 / (0.0091 + 0.3671)$
$LO_{t_1}^{mrkII}$	0.9838	$(Pr(LO_{t_1}(S_1^{mrkII}) .) + Pr(LO_{t_1}(B_1^{mrkII}) .) = (0.4919 + 0.4919)$
$LO_{t_2}^{mrkII}$	0.0080	$(2 \times Pr(LO_{t_1}(S_1^{mrkII}) .) \times 0 + (2 \times Pr(LO_{t_1}(S_2^{mrkI}) .) \times (Pr(LO_{t_2}(S_1^{mrkII}) 0000 - 0100) + Pr(LO_{t_2}(S_2^{mrkII}) 0000 - 0100))) =$ $= (2 \times 0.4919) \times 0 + (2 \times 0.0081) \times (0.0055 + 0.4893)$
LO^{mrkII}	0.4959	$(LO_{t_1}^{mrkII} + LO_{t_2}^{mrkII})/2 = (0.9838 + 0.0080)/2$
$MO_{t_2}^{mrkII}$	0.4794	$(2 \times Pr(LO_{t_1}(S_1^{mrkII}) .) \times Pr(MO_{t_2}(S_1^{mrkII}) 0000 - 0100) + (2 \times Pr(LO_{t_1}(S_2^{mrkI}) .) \times 0 =$ $= (2 \times 0.4919) \times 0.4873 + (2 \times 0.0081) \times 0$
$MO_{t_3}^{mrkII}$	0.2548	$(2 \times Pr(LO_{t_1}(S_1^{mrkII}) .) \times [(Pr(NT_{t_2}(0) 0000 - 0100) + Pr(LO_{t_2}(S_2^{mrkI}) 0000 - 0100)) \times Pr(MO_{t_3}(S_1^{mrkII}) 0000 - 0100)] + 2 \times Pr(LO_{t_1}(S_2^{mrkI}) .)$ $\times (Pr(LO_{t_2}(S_1^{mrkII}) 1000 - 0000) \times Pr(MO_{t_3}(S_1^{mrkII}) 1000 - 0100) + Pr(LO_{t_2}(S_2^{mrkII}) 1000 - 0000) \times Pr(MO_{t_3}(S_2^{mrkII}) 1000 - 1000)) =$ $= 2 \times 0.4919 \times [(0.4920 + 0.0207) \times 0.4975] + (2 \times 0.0081) \times (0.0055 \times 0.4975 + 0.4893 \times 0.4925)$
MO^{mrkII}	0.3671	$(MO_{t_2}^{mrkII} + MO_{t_3}^{mrkII})/2 = (0.4794 + 0.2548)/2$
MS^{mrkII}	0.9758	$MO^{mrkII} / (MO^{mrkI} + MO^{mrkII}) = 0.3671 / (0.0091 + 0.3671)$

Table A5: **Equilibrium Order Submission Strategies, Order Flows and Market Quality**

This table shows how to obtain the metrics on order flows and market quality (column 2) presented in Tables 3 and A7 for the following set of trading fees: $\{MF, TF\} = \{-.001, .001\}$ and $\{mf, tf\} = \{.00, .00\}$. Column 2 reports both the equilibrium market quality metrics for periods t_1 and t_2 , $Spread_{t_z}^j$ and $BBODepth_{t_z}^j$, and the equilibrium average market quality metrics, $Spread^j$ and $BBODepth^j$. Column 3 shows how the values reported in column 2 are computed from the equilibrium strategies. Results are reported for both Market I (mrkI) and Market II (mrkII). Traders have a personal evaluation of the asset $\gamma \sim U[\underline{\gamma}, \bar{\gamma}]$, $S = [0, 2]$ and $AV = 1$. We assume that when the book is empty the quoted spread is equal to 5 ticks, i.e., 0.05. To economize space we indicate the empty book, 0000 – 0000, as ‘|.’

Metric	Value	Analytical Computation
$Spread_{t_1}^{mrkI}$	0.0498	$(2 \times Pr(LO_{t_1}(S_2^{mrkI}) .)) \times 0.04 + (1 - 2 \times Pr(LO_{t_1}(S_2^{mrkI}) .)) \times 0.05 = (2 \times 0.0081) \times 0.04 + (1 - 2 \times 0.0081) \times 0.05$
$Spread_{t_2}^{mrkI}$	0.0497	$(2 \times Pr(LO_{t_1}(S_2^{mrkI}) .)) \times [Pr(LO_{t_2}(S_1^{mrkI}) 1000 - 0000) \times 0.03 + Pr(LO_{t_2}(B_2^{mrkI}) 1000 - 0000) \times 0.03 + Pr(MO_{t_2}(S_2^{mrkI}) 1000 - 0000) \times 0.05 + (1 - (Pr(LO_{t_2}(S_1^{mrkI}) 1000 - 0000) + Pr(LO_{t_2}(B_2^{mrkI}) 1000 - 0000) + Pr(MO_{t_2}(S_2^{mrkI}) 1000 - 0000))) \times 0.04] + 2 \times Pr(LO_{t_2}(S_1^{mrkII}) .) \times [Pr(LO_{t_2}(B_2^{mrkI}) 0000 - 0100) \times 0.04 + (1 - Pr(LO_{t_2}(B_2^{mrkI}) 0000 - 0100)) \times 0.05]$ $= (2 \times 0.0081) \times [0.005 \times 0.03 + 0.0237 \times 0.03 + 0.4765 \times 0.05 + (1 - 0.005 - 0.0237 - 0.4765) \times 0.04] + (2 \times 0.4919) \times [0.0207 \times 0.04 + (1 - 0.0207) \times 0.05]$
$Spread^{mrkI}$	0.0497	$(Spread_{t_1}^{mrkI} + Spread_{t_2}^{mrkI})/2 = (0.0498 + 0.0497)/2$
$Spread_{t_1}^{mrkII}$	0.0303	$(2 \times Pr(LO_{t_1}(S_1^{mrkII}) .)) \times 0.03 + (1 - 2 \times Pr(LO_{t_1}(S_1^{mrkII}) .)) \times 0.05 = 2 \times 0.4919 \times 0.03 + (1 - 2 \times 0.4919) \times 0.05$
$Spread_{t_2}^{mrkII}$	0.0398	$(2 \times Pr(LO_{t_1}(S_1^{mrkII}) .)) \times [Pr(MO_{t_2}(S_1^{mrkII}) 0000 - 0100) \times 0.05 + (1 - Pr(MO_{t_2}(S_1^{mrkII}) 0000 - 0100)) \times 0.03] + (2 \times Pr(LO_{t_1}(S_2^{mrkI}) .) \times [Pr(LO_{t_2}(S_1^{mrkII}) 1000 - 0000) \times 0.03 + Pr(LO_{t_2}(S_2^{mrkII}) 1000 - 0000) \times 0.04 + (1 - Pr(LO_{t_2}(S_1^{mrkII}) 1000 - 0000) - Pr(LO_{t_2}(S_2^{mrkII}) 1000 - 0000)) \times 0.05]$ $= (2 \times 0.4919) \times [0.4873 \times 0.05 + (1 - 0.4873) \times 0.03] + (2 \times 0.0081) \times [0.0055 \times 0.03 + 0.4893 \times 0.04 + (1 - 0.0055 - 0.4893) \times 0.05]$
$Spread^{mrkII}$	0.0351	$(Spread_{t_1}^{mrkII} + Spread_{t_2}^{mrkII})/2 = (0.0303 + 0.0398)/2$
$BBODepth_{t_1}^{mrkI}$	0.0162	$(2 \times Pr(LO_{t_1}(S_2^{mrkI}) .)) \times 1 + (2 \times Pr(LO_{t_1}(S_1^{mrkII}) .)) \times 0 = (2 \times 0.0081) \times 1 + (2 \times 0.4919) \times 0$
$BBODepth_{t_2}^{mrkI}$	0.0292	$(2 \times Pr(LO_{t_1}(S_2^{mrkI}) .)) \times [Pr(LO_{t_2}(S_1^{mrkI}) 1000 - 0000) \times 1 + Pr(LO_{t_2}(B_2^{mrkI}) 1000 - 0000) \times 2 + Pr(MO_{t_2}(S_2^{mrkI}) 1000 - 0000) \times 0 + (1 - (Pr(LO_{t_2}(S_1^{mrkI}) 1000 - 0000) + Pr(LO_{t_2}(B_2^{mrkI}) 1000 - 0000) + Pr(MO_{t_2}(S_2^{mrkI}) 1000 - 0000))) \times 1] + 2 \times Pr(LO_{t_2}(S_1^{mrkII}) .) \times [Pr(LO_{t_2}(B_2^{mrkI}) 0000 - 0100) \times 1 + (1 - Pr(LO_{t_2}(B_2^{mrkI}) 0000 - 0100)) \times 0]$ $= (2 \times 0.0081) \times [0.005 \times 1 + 0.0237 \times 2 + 0.4765 \times 0 + (1 - 0.005 - 0.0237 - 0.4765) \times 1] + (2 \times 0.4919) \times [0.0207 \times 1 + (1 - 0.0207) \times 0]$
$BBODepth^{mrkI}$	0.0227	$(BBODepth_{t_1}^{mrkI} + BBODepth_{t_2}^{mrkI})/2 = (0.0162 + 0.0292)/2$
$BBODepth_{t_1}^{mrkII}$	0.9838	$(2 \times Pr(LO_{t_1}(S_2^{mrkI}) .)) \times 0 + (2 \times Pr(LO_{t_1}(S_1^{mrkII}) .)) \times 1 = (2 \times 0.0081) \times 0 + (2 \times 0.4919) \times 1$
$BBODepth_{t_2}^{mrkII}$	0.5124	$(2 \times Pr(LO_{t_1}(S_1^{mrkII}) .)) \times [Pr(MO_{t_2}(S_1^{mrkII}) 0000 - 0100) \times 0 + (1 - Pr(MO_{t_2}(S_1^{mrkII}) 0000 - 0100)) \times 1] + (2 \times Pr(LO_{t_1}(S_2^{mrkI}) .) \times [Pr(LO_{t_2}(S_1^{mrkII}) 1000 - 0000) \times 1 + (1 - Pr(LO_{t_2}(S_1^{mrkII}) 1000 - 0000) - Pr(LO_{t_2}(S_2^{mrkII}) 1000 - 0000)) \times 0]$ $= (2 \times 0.4919) \times [0.4873 \times 0 + (1 - 0.4873) \times 1] + (2 \times 0.0081) \times [0.0055 \times 1 + 0.4893 \times 1 + (1 - 0.0055 - 0.4893) \times 0]$
$BBODepth^{mrkII}$	0.7481	$(BBODepth_{t_1}^{mrkII} + BBODepth_{t_2}^{mrkII})/2 = (0.9838 + 0.5124)/2$

Table A6: **Equilibrium Order Submission Strategies and Market Quality in Market II. Change in MF of Market I only (for Market II $tf=mf=0.000$). 3-period vs. 4-period Model.**

This table reports for Market II the average equilibrium probabilities of the following order flows and market quality metrics: limit orders, $LO(P_i)$, and market orders, $MO(P_i)$, with the orders breakdown for the outside (P_2) and inside (P_1) price levels - for one side of the market, the other side being symmetric, Quoted Spread (*Quoted Spread*), Best Bid and Offer depth (*BBODepth*), and market share (*MS*). The table reports results obtained under two protocols (models): with 3 periods (Panel A) and 4 periods (Panel B). TF and MF for Market I are reported in rows 1 and 2 (for Market II $\{mf, tf\} = \{.000, .000\}$). Results are reported for different values of MF, holding $TF = 0.000$, specifically: for $MF = 0.000$ (column 1), for $MF = -0.001$ (column 2), and for $MF = -0.005$ (column 4). Column 3 reports the percentage change ($\Delta\%$) in the market quality metrics between $MF = 0.000$ and $MF = -0.001$ and column 5 reports the percentage change ($\Delta\%$) in the market quality metrics between $MF = -0.001$ and $MF = -0.005$. The metrics are reported as averages across all periods of the trading game for each of the two protocols (3 and 4 periods, respectively). For both protocols, the asset value $AV = 1$ and tick size $\tau = 0.01$.

Market II ($tf=mf=0.000$)					
Market I TF	0.000	0.000	$\Delta\%$	0.000	$\Delta\%$
Market I MF	0.000	-0.001	(2) and (1)	-0.005	(4) and (2)
	(1)	(2)	(3)	(4)	(5)
Panel A. 3-period model					
<i>LO</i>	0.2521	0.1599	-36.57%	0.1602	0.16%
<i>LO(P₂)</i>	0.0083	0.0000	-100.00%	0.0000	0.00%
<i>LO(P₁)</i>	0.2438	0.1599	-34.42%	0.1602	0.16%
<i>MO (Volume)</i>	0.1251	0.0398	-68.20%	0.0398	0.16%
<i>MO(P₂)</i>	0.0041	0.0000	-100.00%	0.0000	0.00%
<i>MO(P₁)</i>	0.1211	0.0398	-67.14%	0.0398	0.16%
<i>Quoted Spread</i>	0.0409	0.0444	8.60%	0.0444	-0.02%
<i>BBODepth</i>	0.4624	0.2800	-39.44%	0.2805	0.16%
<i>MS</i>	0.5000	0.1589	-68.21%	0.1591	0.08%
Panel B. 4-period model					
<i>LO</i>	0.2526	0.1216	-51.84%	0.1216	-0.05%
<i>LO(P₂)</i>	0.0682	0.0000	-99.94%	0.0000	-100.00%
<i>LO(P₁)</i>	0.1844	0.1216	-34.05%	0.1216	-0.01%
<i>MO (Volume)</i>	0.1556	0.0592	-61.95%	0.0592	-0.00%
<i>MO(P₂)</i>	0.0352	0.0000	-99.97%	0.0000	-100.00%
<i>MO(P₁)</i>	0.1204	0.0592	-50.79%	0.0592	-0.10%
<i>Quoted Spread</i>	0.0408	0.0445	9.01%	0.0445	-0.01%
<i>BBODepth</i>	0.5106	0.2740	-46.33%	0.2743	0.09%
<i>MS</i>	0.5000	0.3295	-34.10%	0.3297	0.07%

Table A7: **Equilibrium Order Submission Strategies and Market Quality in Market II. Change in MF and TF of Market I only (for Market II $tf=mf=0.000$). 3-period vs. 4-period Model**

This table reports for Market II the average equilibrium probabilities of the following order flows and market quality metrics: limit orders, $LO(P_i)$, and market orders, $MO(P_i)$, with the orders breakdown for the outside (P_2) and inside (P_1) price levels - for one side of the market, the other side being symmetric, Quoted Spread (*Quoted Spread*), Best Bid and Offer depth (*BBO Depth*), and market share (*MS*). The table reports results obtained under two protocols (models): with 3 periods (Panel A) and 4 periods (Panel B). TF and MF for Market I are reported in rows 1 and 2 (for Market II $tf=mf=0.000$). Results are reported for different values of MF, specifically: for $\{MF, TF\} = \{.000, .000\}$ (column 1), for $\{MF, TF\} = \{-.001, .001\}$ (column 2), and for $\{MF, TF\} = \{-.001, .002\}$ (column 4). Column 3 reports the percentage change ($\Delta\%$) in the market quality metrics between $\{MF, TF\} = \{.000, .000\}$ and $\{MF, TF\} = \{-.001, .001\}$ and column 5 reports the percentage change ($\Delta\%$) in the market quality metrics between $\{MF, TF\} = \{.000, .000\}$ and $\{MF, TF\} = \{-.001, .002\}$. The metrics are reported as averages across all periods of the trading game for each of the two protocols (3 and 4 periods, respectively). For both protocols, the asset value $AV = 1$ and tick size $\tau = 0.01$.

Market I TF Market I MF	Market II ($tf=mf=0.000$)				
	0.000 0.000 (1)	0.001 -0.001 (2)	$\Delta\%$ (2) and (1) (3)	0.002 -0.001 (4)	$\Delta\%$ (4) and (1) (5)
Panel A. 3-period model					
<i>LO</i>	0.2521	0.3306	31.15%	0.3307	31.17%
<i>LO(P₂)</i>	0.0083	0.0026	-68.14%	0.0026	-68.24%
<i>LO(P₁)</i>	0.2438	0.3280	34.51%	0.3280	34.53%
<i>MO (Volume)</i>	0.1251	0.2448	95.66%	0.2448	95.68%
<i>MO(P₂)</i>	0.0041	0.0013	68.04%	0.0013	-67.13%
<i>MO(P₁)</i>	0.1211	0.2435	101.14%	0.2435	101.16%
<i>Quoted Spread</i>	0.0409	0.0383	-6.20%	0.0383	-6.24%
<i>BBO Depth</i>	0.4624	0.5846	26.41%	0.5846	26.44%
<i>MS</i>	0.5000	0.9759	95.19%	0.9761	95.21%
Panel B. 4-period model					
<i>LO</i>	0.2521	0.2526	0.00%	0.3729	47.65%
<i>LO(P₂)</i>	0.0682	0.0026	-96.18%	0.1208	77.11%
<i>LO(P₁)</i>	0.1844	0.2500	35.57%	0.2521	36.75%
<i>MO (Volume)</i>	0.1556	0.2151	38.24%	0.2589	66.39%
<i>MO(P₂)</i>	0.0352	0.0010	-97.24%	0.0439	24.65%
<i>MO(P₁)</i>	0.1204	0.2141	77.85%	0.2150	78.53%
<i>Quoted Spread</i>	0.0408	0.0403	-1.30%	0.0388	-5.02%
<i>BBO Depth</i>	0.5106	0.4884	-4.35%	0.6399	25.33%
<i>MS</i>	0.5000	0.7368	47.36%	0.8117	62.35%

Appendix 2: ASX Sample Descriptive Statistics

Table A8: **Descriptive Statistics for 2013 Event, ASX Sample.**

This table reports summary statistics for the control group ASX variables. Our 120 ASX listed stocks sample is stratified by price and market capitalization, based on daily averages for the month of January 2012. All variables reported in the tables, daily measures at the stock level, are for the listing exchange only. *Volume* is defined as the daily number of shares (in 000s) at the end-of-day files from Thomson Reuters Tick History (TRTH). *Depth* is defined as the daily average of the intraday quoted depth at the ask-side and the bid-side of each quote respectively. *Spread* is defined as the time-weighted daily average of the intraday difference between the ask price and the bid price of each quote. *%Spread* is defined as the time weighted daily average of the intraday ask price minus the bid price divided by the midquote of each quote. The descriptive statistics for the four measures of market quality are based on daily numbers for each stock in the one-month pre-period (December 2012). We also report *market capitalization* (in £millions) and *price* levels (in £) both variables are daily measures for the month of January 2012. In addition to the overall samples, for all of our variables we also report summary statistics for the subsamples of the highest (*Large*) and lowest (*Small*) market capitalization terciles.

Market Quality Measures		Mean	Median	ST dev	Q1	Q3
Volume (000s)	Large	4,795	4,054	1,778	3,652	5,202
	Small	2,714	2,536	1,044	2,118	2,745
	Overall	3,905	3,553	1,506	3,016	4,228
Depth	Large	60,644	60,506	9,633	54,803	63,221
	Small	114,969	123,049	37,122	81,329	141,860
	Overall	87,779	88,367	16,151	75,053	99,292
Spread	Large	0.019	0.018	0.002	0.018	0.020
	Small	0.014	0.014	0.001	0.013	0.014
	Overall	0.016	0.016	0.001	0.016	0.016
% Spread	Large	0.167%	0.166%	0.009%	0.161%	0.175%
	Small	0.560%	0.567%	0.021%	0.545%	0.576%
	Overall	0.357%	0.358%	0.011%	0.347%	0.367%
Market Capitalization (AUD Mill)	Large	18,540	8,600	23,366	5,296	18,670
	Small	1,050	1,063	168	909	1,178
	Overall	7,290	2,014	15,595	1,183	5,158
Price	Large	15.440	11.450	13.721	4.460	24.000
	Small	3.654	2.640	3.663	1.371	4.703
	Overall	9.172	4.525	11.687	2.620	11.341